

The Topology Is the Sink: Persistent Homology of Causal Attention Graphs Measures Attention Sinks

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Abstract

Persistent homology (PH) of attention graphs is an increasingly popular source of features for detecting hallucinations in large language models. We show that, in decoder-only models, these features are largely a measurement of a much simpler and well-studied object: the attention sink. We introduce the coning defect δ_s of a vertex s and prove a stable reduction theorem: every bar of dimension $k \geq 1$ in the Vietoris–Rips barcode has length at most δ_s , and 0-dimensional total persistence lies within $(N-1)\delta_s$ of the star weight around s . For causal attention with s the first token, δ_0 has a closed form, and $\delta_0 = 0$ exactly when no token attends to another more than both attend to the sink. We check the theorem on 568,028 layer-level attention graphs from 11 settings (5 open-weight models; teacher-forced TruthfulQA and HaluEval, and an on-policy TriviaQA benchmark we build). The bounds are never violated. In every model with a first-token sink, 36–88% of graphs are exactly coned and 0D persistence has a median per-layer rank correlation of at least 0.999 with sink attention; a model without a sink is never coned, as the theorem predicts. 1D persistence is zero on every coned graph, which explains its reported weakness; we also prove that under i.i.d. weights it grows at least linearly in sequence length ($\geq 3N/8 - O(\sqrt{N})$), so its sub-linear growth on real attention is a signature of coning. For detection, sink features recover most of the 0D signal, and neither 0D nor sink-deflated topology adds measurable information to cheap non-topological probes (hidden states, Lookback ratios, log-probabilities) in any setting. We release the extractor, evaluator, and all per-example features.

1 Introduction

A growing family of hallucination detectors treats a transformer’s attention matrix as a weighted graph and summarizes it with persistent homology (PH): barcodes of connected components (H_0) and loops (H_1) along a Vietoris–Rips filtration (Kushnareva et al., 2021; Cherniavskii et al., 2022; Perez & Reinauer, 2022; Kostenok et al., 2023; Bazarova et al., 2026; Samaga et al., 2026). The appeal is that topology might capture global structure in how information flows that token-level statistics miss. A separate line of work has established that decoder-only language models route a large share of attention to the first token, the attention sink (Xiao et al., 2023; Sun et al., 2024; Cancedda, 2024; Gu et al., 2025; Barbero et al., 2025). These two literatures do not cite each other.

We show they study, to a large and quantifiable extent, the same thing. When one vertex s is closer to every other vertex than those vertices are to each other, every sublevel set of the Vietoris–Rips filtration is a cone with apex s : all higher homology vanishes and the minimum spanning tree (which computes H_0) is the star around s . Real attention graphs sit at or very near this configuration, because of the sink. The “topology” of a causal attention graph is therefore, in most layers, exactly a function of how much attention each token sends to the sink, and in the remaining layers is controlled by a single computable number, the coning defect.

Contributions.

1. A stable sink-reduction theorem (§3). For any symmetric dissimilarity and any vertex s , $k \geq 1$ bars have length $\leq \delta_s$ and 0D total persistence is sandwiched between $S_s - (N-1)\delta_s$ and S_s (Theorem 1). For causal attention, δ_0 and S_0 are closed-form functions of attention to the first token (Corollary 1).
2. Verification on real models (§5.1). There are zero bound violations in 568,028 graphs; in sink-dominated models 36–88% of graphs are exactly coned and H_1 is empty on every coned graph, while a model without a sink serves as a negative control.
3. An account of 1D persistence (§5.3). We prove $\mathbb{E}[P_1(N)] \geq 3N/8 - O(\sqrt{N})$ under i.i.d. weights (Proposition 1), show real attention scales far more slowly, and trace the gap to coning with a sink dose-response experiment in which a planted cycle is visible to H_1 only if it breaks the cone.
4. A detection audit with strong baselines (§5.4). On TruthfulQA, HaluEval, and an on-policy TriviaQA benchmark we build, we compare 0D/1D PH, sink features, and sink-deflated PH against length, lexical, log-probability, LLM-Check (Sriramanan et al., 2024), Lookback-Lens (Chuang et al., 2024), and hidden-state probes, with repeated grouped cross-validation and equivalence tests.

2 Background and related work

Topology of attention. Kushnareva et al. (2021) and Cherniavskii et al. (2022) introduced barcodes of symmetrized BERT attention graphs as features for artificial-text detection and acceptability; Perez & Reinauer (2022) and Kostenok et al. (2023) extended them to classification and uncertainty estimation. For hallucination detection, TOHA (Bazarova et al., 2026) computes an MST-based divergence between prompt and response subgraphs, HalluZig (Samaga et al., 2026) uses zigzag persistence across layers, and CHARM (Frasca et al., 2026) learns a message-passing model over attention graphs. Gardinazzi et al. (2024) study zigzag persistence of hidden representations. All 0D summaries in this family are MST statistics (Fact 1; Gower & Ross, 1969); our results concern what those statistics measure in causal models.

Attention sinks. Xiao et al. (2023) observed that streaming LMs keep the first tokens’ key-values because a large share of attention lands there; Sun et al. (2024) tied sinks to massive activations, Cancedda (2024) to low-rank “dark signals”, Gu et al. (2025) characterized when they emerge during pre-training, and Barbero et al. (2025) argued they prevent over-mixing. Our theorem turns sink strength into an exact statement about the topology of attention graphs.

Hallucination detection. Beyond topology, detectors use sampling consistency (Manakul et al., 2023; Farquhar et al., 2024), internal probes (Azaria & Mitchell, 2023; Burns et al., 2022; Marks & Tegmark, 2023; Li et al., 2023b; Kossen et al., 2024; Orgad et al., 2024; Du et al., 2024), covariance spectra (Chen et al., 2024), attention-kernel log-determinants (Sriramanan et al., 2024), and context-vs-generated attention ratios (Chuang et al., 2024). Recent audits find that several benchmarks admit near-perfect detection from surface form alone (Hussain & Kantarcioglu, 2026; Janiak et al., 2025), echoing annotation artifacts in NLI (Gururangan et al., 2018; McCoy et al., 2019); we therefore include lexical controls and an on-policy benchmark.

3 Theory: attention sinks cone the Vietoris–Rips filtration

Let $D \in \mathbb{R}_{\geq 0}^{N \times N}$ be symmetric with zero diagonal. The Vietoris–Rips filtration is $K_\epsilon = \{\sigma \subseteq [N] : \max_{u,v \in \sigma} D_{uv} \leq \epsilon\}$; $\text{dgm}_k(D)$ is the barcode of $H_k(K_\epsilon; \mathbb{F})$ over a field, and $P_k(D)$ is its total persistence (sum of finite bar lengths). For attention, $A^{(l)}$ is the head-averaged, row-stochastic, lower-triangular attention matrix of layer l and, following the literature, $D = 1 - \max(A, A^\top)$ with zero diagonal.

Fact 1 (Gower & Ross, 1969). All 0D bars are born at 0, and their finite death times are, as a multiset, the edge weights of a minimum spanning tree (MST) of D . In particular $P_0(D) = w(\text{MST}(D))$.

Definition 1 (Coning defect, star weight). For $s \in [N]$, the coning defect is $\delta_s(D) = \max_{u \neq v, u, v \neq s} [\max(D_{su}, D_{sv}) - D_{uv}]_+$ (zero if $N < 3$), and the star weight is $S_s(D) = \sum_{u \neq s} D_{su}$. We call D coned at s if $\delta_s(D) = 0$, i.e. s is at least as close to both endpoints of every edge as they are to each other.

Theorem 1 (Sink-coning reduction). For every symmetric D and every vertex s , with $\delta = \delta_s(D)$:

- (a) every bar of $\text{dgm}_k(D)$, $k \geq 1$, has length at most δ ; hence $P_k(D) \leq \delta \cdot |\text{dgm}_k(D)|$;
- (b) $S_s(D) - (N-1)\delta \leq P_0(D) \leq S_s(D)$, and $\max_u D_{su} - \delta \leq \max \text{dgm}_0^{\text{fin}}(D) \leq \max_u D_{su}$;
- (c) if D is coned at s , then $H_k(K_\epsilon) = 0$ for all $k \geq 1$ and all ϵ , the star at s is an MST, and $P_0(D) = S_s(D)$.

Proof sketch (full proof in Appendix A). Let $D'_{uv} = \max(D_{uv}, D_{su}, D_{sv})$ for $u, v \neq s$ and $D'_{su} = D_{su}$. Then $D \leq D' \leq D + \delta$ entrywise and D' is coned at s . If D' is coned, any simplex $\sigma \not\ni s$ with $|\sigma| \geq 2$ has the same filtration value as $\sigma \cup \{s\}$, and a vertex not yet joined to s has no edges; so each K'_ϵ is a cone with apex s plus isolated points, and $H_k(K'_\epsilon) = 0$ for $k \geq 1$. The inclusions $K_b \subseteq K'_{b+\delta} \subseteq K_{b+\delta}$ factor the persistence map $H_k(K_b) \rightarrow H_k(K_{b+\delta})$ through zero, so no bar outlives δ . For (b), the star is a spanning tree of D (upper bound) and an MST of D' by the cycle property; MST weight and bottleneck are $(N-1)$ - and 1-Lipschitz in the sup-norm. $\square$

The bound in (a) is tight up to the constant: on 3,000 random causal matrices the largest observed ratio of H_1 lifetime to δ_0 was 0.99.

Corollary 1 (Causal attention). Let A be lower-triangular and row-stochastic, $D = 1 - \max(A, A^\top)$, and $s = 0$ the first token. Then $D_{0u} = 1 - A_{u0}$, so $S_0 = (N-1)(1 - \bar{m})$ with $\bar{m} = \frac{1}{N-1} \sum_{u \geq 1} A_{u0}$ the mean sink attention, and

$$\delta_0 = \max_{1 \leq v < u} [A_{uv} - \min(A_{u0}, A_{v0})]_+.$$

Thus $P_0 \in [(N-1)(1 - \bar{m} - \delta_0), (N-1)(1 - \bar{m})]$ and all $H_{k \geq 1}$ bars are shorter than δ_0 . The graph is exactly coned iff no token attends to an earlier non-sink token more than both of them attend to the sink.

Remark 1 (Choice of distance). Coning is an order condition, so it is invariant under any entrywise non-decreasing transform of D (e.g. $-\log A$ or ranks in place of $1 - A$): the set of exactly coned graphs, and the vanishing of their higher homology, does not depend on how attention is turned into a distance. Only the magnitude of δ_s on non-coned graphs does.

Corollary 1 makes four testable predictions: (P1) in sink-dominated layers 0D persistence is an affine function of sink attention; (P2) H_1 is empty whenever $\delta_0 = 0$; (P3) 1D persistence can only be non-zero where the sink is violated; (P4) removing the sink and renormalizing (deflation) exposes whatever non-sink topology exists.

What i.i.d. weights would give. Earlier work motivates length normalization of 1D features by the growth of cycles with N . That growth is real, but only for graphs far from coned.

Proposition 1 (Linear growth under i.i.d. weights). If $\{D_{uv}\}_{u < v}$ are i.i.d. Uniform $[0, 1]$, then $\mathbb{E}[P_1(N)] \geq \int_0^1 \left[\binom{N}{2} p - N - \binom{N}{3} p^3 \right]_+ dp = \frac{3}{8}N - O(\sqrt{N})$, and $\mathbb{E}[P_1(N)] \leq \binom{N}{2}/2$.

Proof sketch. K_p is the clique complex of $G(N, p)$ (Kahle, 2009) and $P_1 = \int_0^1 \beta_1(K_p) dp$. The strong Morse inequality (Edelsbrunner & Harer, 2010) gives $\beta_1 \geq f_1 - f_0 - f_2$; take expectations, use $\beta_1 \geq 0$, and integrate with $p = x/\sqrt{N}$. The upper bound uses $\beta_1 \leq f_1$. Details in Appendix A. $\square$

Table 1: Theorem 1 on real attention. cells: (example, layer) graphs; viol.: violations of Theorem 1(a) or (b) with $s = 0$; coned: share with $\delta_0 = 0$ exactly; $H_1=0$ | coned: share of coned graphs with empty H_1 (P2 predicts 100%); last column: median over layers of Spearman(P_0, S_0) across examples.

Benchmark	Model	$\bar{N}$	graphs	viol.	sink attn.	coned	$H_1=0$ coned	$\rho(P_0, S_0)$
TruthfulQA	Qwen2.5-3B	44.2	58,824	0	0.50	62.2%	100.0%	1.000
TruthfulQA	Qwen2.5-1.5B	44.2	45,752	0	0.53	64.5%	100.0%	1.000
TruthfulQA	Phi-3-mini	29.0	52,288	0	0.69	88.4%	100.0%	1.000
TruthfulQA	TinyLlama-1.1B	42.3	35,948	0	0.53	36.2%	100.0%	1.000
TruthfulQA	SmolLM-1.7B	32.4	39,216	0	0.02	0.0%	–	0.967
HaluEval	Qwen2.5-3B	126.9	72,000	0	0.42	53.0%	100.0%	1.000
HaluEval	Qwen2.5-1.5B	128.1	28,000	0	0.45	62.5%	100.0%	1.000
TriviaQA (on-policy)	Qwen2.5-3B	68.6	72,000	0	0.48	63.0%	100.0%	1.000
TriviaQA (on-policy)	Qwen2.5-1.5B	68.8	56,000	0	0.52	50.6%	100.0%	1.000
TriviaQA (on-policy)	Phi-3-mini	46.1	64,000	0	0.69	87.1%	100.0%	1.000
TriviaQA (on-policy)	TinyLlama-1.1B	72.0	44,000	0	0.50	36.4%	100.0%	1.000

Directed complexes do not escape the reduction. Symmetrization is often blamed for discarding direction. For causal attention it does not: the directed flag complex with weights $1 - A_{ij}$ is isomorphic, as a filtered complex, to $\text{VR}(D)$, because token order is a total order on the support of A (Theorem 2, Appendix A.4). Theorem 1 therefore applies to it verbatim.

4 Experimental setup

Models and benchmarks. We use Qwen2.5-1.5B/3B-Instruct, Phi-3-mini-4k-instruct, TinyLlama-1.1B-Chat, and SmolLM-1.7B-Instruct (Mistral-7B-Instruct results appear in Appendix D when available). TruthfulQA (Lin et al., 2022): all 817 questions, each paired with its best answer and first incorrect answer (1,634 rows) under each model’s native chat format. HaluEval QA (Li et al., 2023a): the first 1,000 (Qwen2.5-3B) or 500 (Qwen2.5-1.5B) knowledge-grounded questions with right and hallucinated answers. Both are teacher-forced: the model reads dataset-written answers. TriviaQA on-policy: 2,000 random validation questions (Joshi et al., 2017); each model answers greedily (short-phrase instruction, ≤ 24 tokens), an answer is labelled hallucinated unless a normalized gold alias occurs in it, and features are extracted on the model’s own answer. Per-model accuracies are in Table 4.

Features. Per layer (head-averaged A): 0D (max lifetime, total persistence); 1D (four summaries, divided by N); Sink (S_0 and $\max_u D_{0u}$, i.e. exactly the two quantities Corollary 1 says 0D reduces to); Deflated PH (0D+1D after deleting token 0 and renormalizing rows); Answer-only PH (induced subgraph on answer tokens); row entropy and row max. Per head: 0D (MST total and max via a dense Prim’s algorithm), sink mass, entropy, Lookback ratio (Chuang et al., 2024), and the LLM-Check attention score (Sriramanan et al., 2024). Other: answer-token log-probability and predictive-entropy statistics; mean-pooled answer hidden states at the middle layer; TF-IDF word and character n -grams of the answer text (Lexical).

Protocol. Every number uses one evaluator: standardized features, L_2 logistic regression ($C=1$ for ≤ 64 dimensions; otherwise $C \in \{0.01, 0.1, 1\}$ by inner grouped 3-fold CV), 10-fold StratifiedGroupKFold grouped by question, repeated over 3 seeds with out-of-fold probabilities averaged. Differences in pooled AUC use a 10,000-resample question-level cluster bootstrap; we call two banks equivalent if the 90% interval lies inside ± 0.015 (TOST at $\alpha=0.05$) and report TOST power. Paired benchmarks also report within-pair accuracy. The theory checks use every (example, layer) graph; ripser (Bauer, 2021) stores filtration values in 32-bit floats, so bounds are checked at tolerance 10^{-5} .

5 Results

5.1 The reduction holds on real attention

Table 1 checks Theorem 1 on every layer-level graph of every example: 568,028 graphs in total. There are 0 violations of either bound. In the 10 settings whose models place

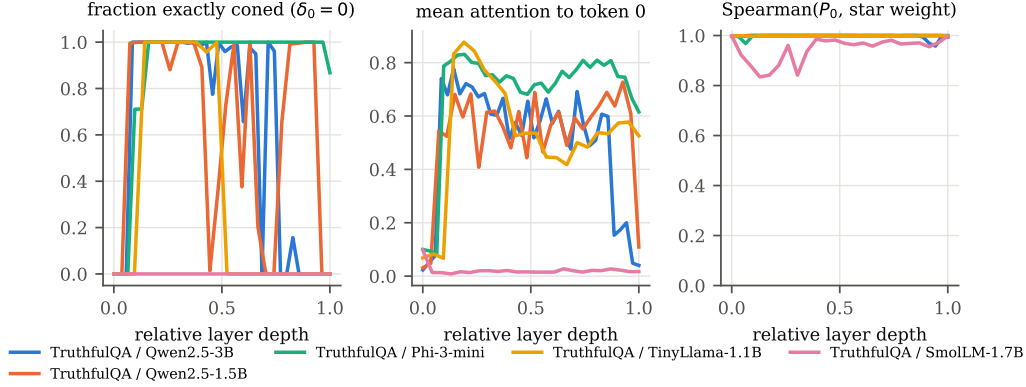


Figure 1: Layerwise structure (TruthfulQA). Coning (left) tracks sink attention (middle): layers with little sink attention, typically the first and last few, are rarely coned, and SmolLM-1.7B, which has no first-token sink, is never coned. Even there, P_0 remains rank-aligned with the star weight (right), as Theorem 1(b) permits when δ_0 is small relative to the spread of S_0 .

0.42 to 0.69 of attention on token 0, 36–88% of graphs are exactly coned ($\delta_0 = 0$), and on every one of them H_1 is empty, as prediction P2 requires. Across graphs, δ_0 and 1D total persistence have Spearman correlation 0.33 to 0.89 (P3). The median per-layer rank correlation between P_0 and the star weight S_0 is at least 0.999 in every sink setting (P1); since $S_0 = (N-1)(1 - \bar{m})$, 0D total persistence is, up to rank, the mean attention paid to the first token.

Figure 1 shows where coning happens: in the middle of the network, where sink attention is high. The first few layers are never sink-dominated, and in some models (TinyLlama, the Qwen models) so is a block of late layers; these layers are rarely coned. There, P_0 still tracks S_0 closely, because δ_0 is small relative to the variation of S_0 across examples.

A model without a sink is a natural control. SmolLM-1.7B, under its ChatML format, places only 0.02 of its attention on token 0, and no vertex cones any of its graphs (the best apex never reaches $\delta_s = 0$). Theorem 1 then predicts no reduction, and indeed its H_1 is non-empty on most graphs and its per-layer $\rho(P_0, S_0)$ drops to 0.967. The reduction is a property of sink-dominated attention, not an artifact of the pipeline.

5.2 What topology adds beyond the sink

If 0D persistence reduced exactly to the star, a classifier on the two sink features would match one on the two 0D features in every layer. Table 3 (“Sink vs. 0D”) shows the two banks differ by -0.015 to +0.031 AUC across the 10 sink settings evaluated, and are statistically equivalent within ± 0.015 in 4 of them. Adding 0D to the sink bank gains +0.004 to +0.030.

The residual lives exactly where the theory says it can. We split layers without using labels: coned layers ($\geq 90\%$ of graphs coned) and open layers ($< 50\%$). Adding the coned layers’ 0D features to Sink changes AUC by -0.001 to +0.004, whereas the open layers account for +0.003 to +0.029 (Appendix C). Almost everything 0D adds beyond sink attention comes from the layers where the sink does not dominate.

Per head, the reduction is tighter. Per-head sink mass and per-head 0D are far stronger than their head-averaged versions (Table 2), but adding per-head 0D to per-head sink mass changes AUC by only -0.001 to +0.011, which is equivalent within ± 0.015 in 9 of 9 settings.

Deleting the sink exposes real, but redundant, structure. Deflated PH (P4) adds +0.032 to +0.045 over Sink on TruthfulQA and +0.002 to +0.029 on the on-policy benchmark, and answer-only PH behaves similarly (Appendix C). So attention graphs carry some non-sink

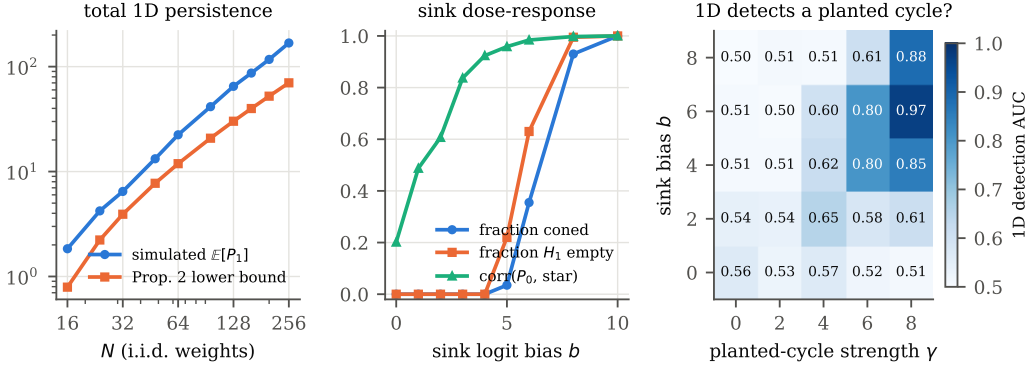


Figure 2: Synthetic checks. Left: Proposition 1’s lower bound against the simulated mean total 1D persistence under i.i.d. weights (8 draws per N). Middle: random causal attention ($N=40$, 200 draws per point) with a logit bias b on token 0: as the sink strengthens, graphs become exactly coned, H_1 empties, and P_0 becomes a function of the star weight. Right: AUC of 1D total persistence for detecting a planted 6-cycle of logit strength γ (150 draws per class). Under a strong sink ($b=8$) a planted cycle is invisible unless it is strong enough to break the cone.

geometry. As §5.4 shows, however, this geometry is already available to non-topological probes.

5.3 One-dimensional persistence

Under i.i.d. weights, 1D persistence grows super-linearly over the range that matters, and Proposition 1 guarantees at least linear growth asymptotically (Figure 2, left). Real attention behaves very differently: most graphs have no 1D bars at all (Table 1), and the bars that exist are confined to graphs with a non-zero coning defect. Figure 2 (middle) reproduces this transition in a controlled setting: raising the sink’s logit bias moves random causal attention from the i.i.d.-like regime to exact coning, with H_1 emptying in lockstep.

This explains a recurring empirical observation, that 1D features add little to 0D. At full sample size, adding length-normalized 1D to 0D changes AUC by -0.008 to +0.026 and is equivalent within ± 0.015 in 8 of 10 sink settings. The mechanism is not that loops are uninformative in principle. In a strongly sinked model, a loop can only exist where some token pays more attention to another token than both pay to the sink, and its lifetime is bounded by that excess. Figure 2 (right) makes this concrete: a planted 6-cycle that H_1 detects easily under a moderate sink ($b=6$, $\gamma=8$: AUC 0.97) is invisible under a strong one ($b=8$, $\gamma \leq 4$: AUC ≤ 0.51) until it is strong enough to break the cone.

5.4 Hallucination detection against strong baselines

Table 2 places attention-graph topology among standard detectors. Three patterns hold across benchmarks.

Topology is dominated by cheap non-topological probes. On TruthfulQA and the on-policy benchmark, a mean-pooled hidden-state probe (0.83 to 0.93) and per-head Lookback ratios (0.82 to 0.90) outperform every topological bank, and their combination with log-probability, LLM-Check, and row statistics reaches 0.84 to 0.94. Adding 0D PH to that combination changes AUC by 0.000 to +0.001 (equivalent within ± 0.015 in 10 of 10 settings), and adding sink-deflated PH changes it by -0.001 to +0.002 (equivalent in 10 of 10).

Which cheap probe wins depends on whether text is teacher-forced. On TruthfulQA, token log-probability is weak (0.58 to 0.64), consistent with the benchmark’s design around imitative falsehoods, and 0D PH (0.58 to 0.75) beats it for four of five models. On-policy, log-probability is strong (0.72 to 0.88) and beats 0D (0.71 to 0.73) for three of four models,

Table 2: Pooled out-of-fold ROC-AUC (3-seed repeated grouped 10-fold CV; seed SD ≤ 0.008). TQA: TruthfulQA; HE: HaluEval; TrQA: on-policy TriviaQA. Lex.: answer-text TF-IDF; LogP: log-probability statistics; LLM-C.: LLM-Check; Sink: ($S_0, \max_u D_{0u}$) per layer; 0D/1D: persistence per layer; subscript h : per head; Lookb.: Lookback ratios; Hidden: mid-layer probe; Non-topo.: LogP+LLM-C.+Lookb.+Hidden+row statistics. Best per row in bold; “-”: not computed (no per-head features for HE Qwen2.5-3B). Full table in Appendix C.

Setting	Lex.	LogP	LLM-C.	Sink	0D	1D	Sink _h	0D _h	Lookb.	Hidden	Non-topo.
TQA Qw3B	.820	.631	.693	.740	.754	.602	.850	.854	.897	.913	.927
TQA Qw1.5B	.820	.638	.666	.723	.726	.586	.830	.838	.851	.916	.917
TQA Phi-3	.820	.578	.653	.652	.647	.529	.871	.868	.892	.933	.936
TQA TinyLl.	.820	.637	.614	.644	.654	.593	.807	.805	.850	.883	.885
TQA SmolLM	.820	.589	.604	.611	.584	.545	.720	.798	.863	.894	.899
HE Qw3B	.981	.993	-	.827	.816	.709	-	-	-	.999	-
HE Qw1.5B	.979	.993	.791	.820	.789	.747	.922	.949	.997	.999	.999
TrQA Qw3B	.596	.839	.688	.717	.732	.601	.820	.807	.861	.841	.870
TrQA Qw1.5B	.609	.823	.676	.707	.716	.574	.767	.768	.824	.855	.873
TrQA Phi-3	.635	.877	.676	.707	.710	.546	.826	.807	.840	.867	.883
TrQA TinyLl.	.740	.724	.688	.720	.720	.659	.814	.775	.823	.827	.842

Table 3: Key paired comparisons (Δ AUC). “X–Y” is $AUC(X) - AUC(Y)$; “X|Y” is $AUC(Y+X) - AUC(Y)$. Defl.: sink-deflated 0D+1D PH; NT: all non-topological features. $\equiv$: equivalent within ± 0.015 (90% interval inside the margin); *: 95% interval excludes 0. Intervals and TOST power for every cell are in Appendix C.

Setting	Sink–0D	0D Sink	0D _h Sink _h	Defl. Sink	1D 0D	0D NT	Defl. NT
TQA Qw3B	−0.014*	+0.017*	+0.010 $\equiv$	+0.032*	+0.006 $\equiv$	+0.001 $\equiv$	+0.001 $\equiv$
TQA Qw1.5B	−0.003 $\equiv$	+0.012*	+0.011 $\equiv$	+0.034*	−0.008 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
TQA Phi-3	+0.005 $\equiv$	+0.006 $\equiv$	0.000 $\equiv$	+0.033*	−0.001 $\equiv$	0.000 $\equiv$	+0.001 $\equiv$
TQA TinyLl.	−0.010	+0.020*	+0.004 $\equiv$	+0.045*	+0.005 $\equiv$	0.000 $\equiv$	−0.001 $\equiv$
TQA SmolLM	+0.027*	+0.030*	+0.105*	+0.041*	+0.004 $\equiv$	0.000 $\equiv$	0.000 $\equiv$
HE Qw3B	+0.011	+0.030*	−	+0.055*	+0.013*	−	−
HE Qw1.5B	+0.031*	+0.012*	+0.008 $\equiv$	+0.052*	+0.026*	0.000 $\equiv$	0.000 $\equiv$
TrQA Qw3B	−0.015*	+0.021*	+0.004 $\equiv$	+0.002	−0.003 $\equiv$	+0.001 $\equiv$	0.000 $\equiv$
TrQA Qw1.5B	−0.009	+0.010	+0.004 $\equiv$	+0.012	−0.002 $\equiv$	0.000 $\equiv$	−0.001 $\equiv$
TrQA Phi-3	−0.004 $\equiv$	+0.006 $\equiv$	+0.002 $\equiv$	+0.029*	−0.005 $\equiv$	0.000 $\equiv$	+0.002 $\equiv$
TrQA TinyLl.	+0.001 $\equiv$	+0.004 $\equiv$	−0.001 $\equiv$	+0.014	−0.002 $\equiv$	0.000 $\equiv$	0.000 $\equiv$

tying on TinyLlama. Conclusions about topology versus confidence drawn from a single teacher-forced benchmark do not transfer.

Per-head resolution matters more than topology. Per-head features of every kind are far stronger than head-averaged ones, and per-head entropy lands within -0.016 to +0.036 AUC of per-head 0D (Appendix C).

5.5 Benchmark artifacts

A TF-IDF model on the answer text alone reaches 0.82 AUC on TruthfulQA and 0.98 on HaluEval, beating every head-averaged attention feature (on HaluEval even a hidden-state probe reaches 1.00, which says more about the benchmark than the probe). On-policy, where the text is the model’s own, the same model reaches only 0.60 to 0.74. Teacher-forced benchmarks therefore reward detectors for recognizing how the dataset’s true and false answers were written, a problem shared with other hallucination corpora (Hussain & Kantarcioglu, 2026; Janiak et al., 2025). On HaluEval, where lexical separability is near perfect, we report results for completeness (Appendix C) but draw no conclusions from them.

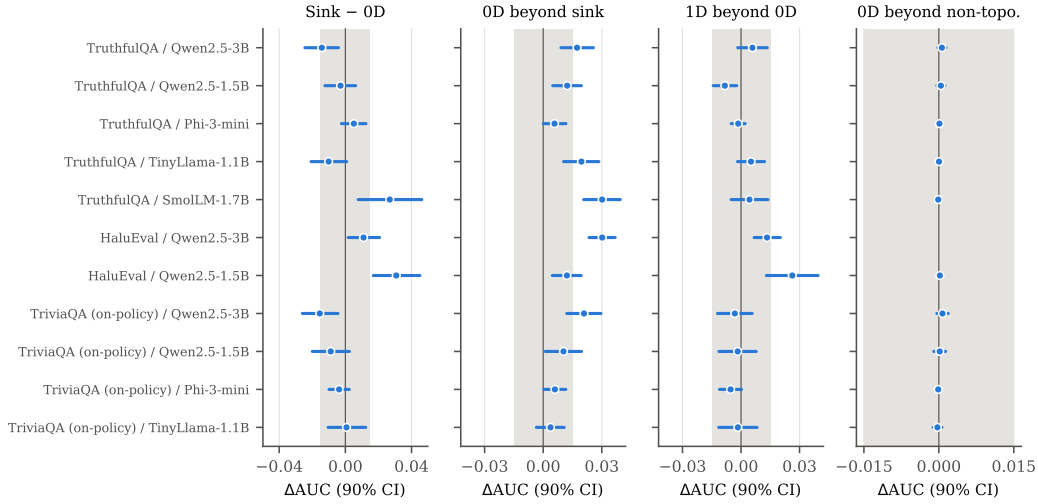


Figure 3: Four comparisons across all settings (ΔAUC with 90% cluster-bootstrap intervals; shaded band: ± 0.015). Sink features nearly reproduce 0D; 0D adds little beyond them; 1D adds nothing beyond 0D; and 0D adds nothing beyond non-topological probes.

6 Discussion

What attention-graph topology measures. In decoder-only models, head-averaged Vietoris–Rips persistence is controlled by one scalar per graph, the coning defect of the first token. Where it is zero, which is the majority of middle-layer graphs in every sink-dominated model we tested, 0D persistence is the sink attention and higher homology is empty. Where it is positive, the bounds of Theorem 1 say how far topology can depart from the sink, and the data show that this departure is concentrated in a few early and late layers.

Recommendations. (i) Report sink attention and δ_0 next to any attention-graph topological feature. Both are cheap ($O(N)$ and $O(N^2)$), and δ_0 says in advance whether higher-dimensional features can be non-zero at all. (ii) Compare topological detectors against per-head non-topological statistics (sink mass, entropy, Lookback ratios) at the same resolution, and against a hidden-state probe. (iii) Evaluate on the model’s own generations; teacher-forced benchmarks mix detection with recognition of dataset authorship. (iv) If structure beyond the sink is the goal, remove the sink first (deflation) or restrict to answer tokens. Both expose real non-sink geometry, but in our settings it was not new information.

Limitations. Our models range from 1.1B to 3.8B parameters (7B in the appendix when available); sink strength changes with scale and training (Gu et al., 2025), so the share of coned layers is model-specific, although Theorem 1 is not. We study the head-averaged symmetrized construction and its causal directed-flag equivalent; zigzag constructions across layers (Samaga et al., 2026) and learned graph models (Frasca et al., 2026) are outside the theorem, though their inputs are the same sink-dominated graphs. Our on-policy benchmark uses string-match grading, which mislabels some paraphrases. Teacher-forced benchmarks measure whether attention separates dataset-written true and false text, which Section 5.5 shows is partly a property of how that text was written.

Reproducibility statement

All code is released at <https://anonymous.4open.science/r/ICLR-BE07/>: sinktda/extract.py (all features), sinktda/evaluate.py (every AUC, interval, and theory check), sinktda/synthetic.py (Figure 2), and sinktda/report.py (every table and figure). Per-example features for all settings and the TriviaQA generations are included. Proofs are in Appendix A.

LLM usage statement

LLMs were used to help implement experiment code, check reported statistics against result files, and draft and edit text. The authors verified every claim, number, proof, and citation and take full responsibility for the paper.

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A Proofs

Throughout, D is a symmetric $N \times N$ matrix with non-negative entries and zero diagonal, K_ϵ is the Vietoris–Rips complex at scale ϵ , $\text{filt}_D(\sigma) = \max_{u,v \in \sigma} D_{uv}$ (with $\text{filt}_D(\{u\}) = 0$), and homology is taken over a field. Bars are half-open intervals $[b, d)$.

A.1 Fact 1

All vertices enter at 0. Processing edges in non-decreasing weight order, an edge either joins two components, in which case (elder rule) one 0D class dies at its weight and Kruskal’s algorithm adds the edge to the spanning forest, or it closes a cycle, in which case no 0D class dies and Kruskal discards it. Hence the $N - 1$ finite death times are the MST edge weights (Gower & Ross, 1969; Edelsbrunner & Harer, 2010).

A.2 Theorem 1

Lemma 1 (Coned filtrations). If D is coned at s , then for every $\epsilon \geq 0$, $K_\epsilon = C_\epsilon \sqcup I_\epsilon$, where C_ϵ is a simplicial cone with apex s and I_ϵ is a set of isolated vertices. Consequently $\tilde{H}_k(K_\epsilon) = 0$ for every $k \geq 1$.

Proof. Let $\sigma \in K_\epsilon$ with $s \notin \sigma$ and $|\sigma| \geq 2$. For each $u \in \sigma$ choose $v \in \sigma \setminus \{u\}$; coning gives $D_{su} \leq D_{uv} \leq \text{filt}_D(\sigma) \leq \epsilon$. Therefore $\text{filt}_D(\sigma \cup \{s\}) = \text{filt}_D(\sigma)$ and $\sigma \cup \{s\} \in K_\epsilon$. Now let $u \neq s$ be a vertex with $D_{su} > \epsilon$. If u had an edge $\{u, v\} \in K_\epsilon$ then $D_{su} \leq D_{uv} \leq \epsilon$, a contradiction; so u is isolated. Let I_ϵ be these vertices and C_ϵ the subcomplex on the remaining vertices. Every vertex of C_ϵ is adjacent to s , and by the first part every simplex of C_ϵ extends by s , so C_ϵ is the cone over its link of s and is contractible. H_k of a disjoint union is the direct sum, and isolated points have no homology in degree ≥ 1 . $\square$

Proof of Theorem 1. Fix s and $\delta = \delta_s(D)$. Define $D'_{uv} = \max(D_{uv}, D_{su}, D_{sv})$ for $u, v \neq s$, $u \neq v$; $D'_{su} = D'_{us} = D_{su}$; $D'_{uu} = 0$.

Step 1: $D \leq D' \leq D + \delta$ and D' is coned at s . The first inequality is immediate. For $u, v \neq s$, $D'_{uv} - D_{uv} = [\max(D_{su}, D_{sv}) - D_{uv}]_+ \leq \delta$. Coning: $\max(D'_{su}, D'_{sv}) = \max(D_{su}, D_{sv}) \leq D'_{uv}$.

Step 2: interleaving. Since filtration values are maxima of entries, $\text{filt}_D \leq \text{filt}_{D'} \leq \text{filt}_D + \delta$. Hence for every b , $K_b \subseteq K'_{b+\delta} \subseteq K_{b+\delta}$, where K' is the filtration of D' .

Step 3: (a). The persistence map $H_k(K_b) \rightarrow H_k(K_{b+\delta})$ is induced by the composite inclusion in Step 2 and therefore factors through $H_k(K'_{b+\delta})$, which is zero for $k \geq 1$ by Lemma 1. The rank of $H_k(K_b) \rightarrow H_k(K_{b'})$ equals the number of bars $[b_i, d_i)$ with $b_i \leq b$ and $d_i > b'$. A bar with $d_i - b_i > \delta$ would contribute to this rank at $b = b_i$, $b' = b_i + \delta$, a contradiction. So every bar has length at most δ , and summing gives the bound on P_k .

Step 4: (b). By Fact 1, $P_0(D) = w_D(T_D)$ for an MST T_D of D . The star $T_s = \{\{s, u\}\}_{u \neq s}$ is a spanning tree, so $P_0(D) \leq w_D(T_s) = S_s(D)$. For the lower bound, T_s is an MST of D' : every non-star edge $\{u, v\}$ closes the cycle s, u, v in which it is a heaviest edge ($D'_{uv} \geq \max(D'_{su}, D'_{sv})$), so by the cycle property it can be excluded. Hence $w_{D'}(T_{D'}) = w_{D'}(T_s) = S_s(D)$, and $P_0(D) = w_D(T_D) \geq w_{D'}(T_D) - (N - 1)\delta \geq w_{D'}(T_{D'}) - (N - 1)\delta = S_s(D) - (N - 1)\delta$. The maximum finite 0D death is the bottleneck value $\min_T \max_{e \in T} w(e)$, which is attained by any MST; it is at most $\max_u D_{su}$ (the star is a spanning tree) and is 1-Lipschitz in the sup-norm, which gives the lower bound because the bottleneck of D' is $\max_u D_{su}$.

Step 5: (c). If $\delta = 0$ then $D' = D$, and the claims follow from Lemma 1 and Step 4. $\square$

Proof of Corollary 1. For $u > v$, $A_{vu} = 0$, so $D_{uv} = 1 - A_{uv}$; in particular $D_{0u} = 1 - A_{u0}$. For $1 \leq v < u$, $\max(D_{0u}, D_{0v}) - D_{uv} = A_{uv} - \min(A_{u0}, A_{v0})$. Substitute into Theorem 1. $\square$

A.3 Proposition 1

Proof. With i.i.d. uniform weights, the edges with $D_{uv} \leq p$ form an Erdős–Rényi graph $G(N, p)$ and K_p is its clique (flag) complex. Every 1D bar is contained in $[0, 1)$, because K_1 is the full simplex. Writing $\beta_1(p)$ for the first Betti number of K_p , the number of bars alive at p is $\beta_1(p)$, so $P_1 = \int_0^1 \beta_1(p) dp$. For any finite complex with f_i faces of dimension i , the strong Morse inequality at degree 2 reads $\beta_2 - \beta_1 + \beta_0 \leq f_2 - f_1 + f_0$ (Edelsbrunner & Harer, 2010), hence $\beta_1 \geq f_1 - f_0 - f_2 + \beta_0 + \beta_2 \geq f_1 - f_0 - f_2$. Since also $\beta_1 \geq 0$, Jensen’s inequality gives $\mathbb{E}\beta_1(p) \geq [\mathbb{E}f_1 - f_0 - \mathbb{E}f_2]_+ = [(\binom{N}{2}p - N - \binom{N}{3}p^3)_+]_+$. Integrating over p gives the first inequality. With $p = x/\sqrt{N}$ the integrand is $N^{3/2}(x/2 - x^3/6) - N + O(N^{1/2})$, positive on $x \in (2/\sqrt{N} \cdot (1 + o(1)), \sqrt{3})$, and $\int_0^1 [\cdot]_+ dp = N \int_0^{\sqrt{3}} (\frac{x}{2} - \frac{x^3}{6}) dx - O(\sqrt{N}) = \frac{3}{8}N - O(\sqrt{N})$. For the upper bound, $\beta_1 \leq f_1$ gives $\mathbb{E}P_1 \leq \int_0^1 \binom{N}{2} p dp = \binom{N}{2}/2$. $\square$

Numerically, the integral equals 0.79, 3.92, 11.9, 30.2, 70.1 at $N = 16, 32, 64, 128, 256$, against simulated means 1.84, 6.46, 22.4, 64.9, 168.0 (Figure 2, left). Both grow super-linearly over this range (log-log slopes 1.55 and 1.61); the lower bound’s slope tends to 1 as $N \rightarrow \infty$.

A.4 Directed flag complexes on causal attention

Theorem 2 (Directed flag collapse). Let A be lower-triangular with $A_{ij} > 0$ for all $j \leq i$ (the diagonal included; self-loops are ignored). Let $\text{DFI}(A)$ be the filtered directed flag complex (Lütgehetmann et al., 2020) of the digraph with an edge $i \rightarrow j$ for every $j < i$ and weight $1 - A_{ij}$, and let $\text{VR}(D)$ be as above with $D = 1 - \max(A, A^\top)$. Then $\text{DFI}(A)$ and $\text{VR}(D)$ are isomorphic filtered complexes, so $\text{dgm}_k(\text{DFI}(A)) = \text{dgm}_k(\text{VR}(D))$ for all k . The same holds for any weighting $w(i \rightarrow j) = \phi(A_{ij})$ with $D_{ij} = \phi(\max(A_{ij}, A_{ji}))$ and ϕ non-increasing.

Proof. A directed k -simplex is an ordered tuple $(v_0, \dots, v_k)$ with an edge $v_a \rightarrow v_b$ for all $a < b$. Edges exist only from later to earlier tokens, so a vertex set S supports exactly one directed simplex, its elements in decreasing order; every other ordering uses a missing forward edge. (A missing edge is not a zero-weight edge; encoding it as weight 0 would admit all orderings, a common implementation error.) This bijection preserves faces. Its filtration value is $\max_{a < b} (1 - A_{v_a v_b})$, and for $u > v$, $D_{uv} = 1 - A_{uv}$, so the undirected value $\max_{u, v \in S} D_{uv}$ is the same number. $\square$

We verified the bijection and the filtration identity by explicit enumeration on 420 random causal matrices ($N \leq 8$, four families including tie-dense and underflowing matrices; 30,060 vertex sets; zero violations), and confirmed that the identity fails for bidirectional attention, where a set of size n supports $n!$ orderings. The collapse is a consequence of causality, so directional treatments that are well motivated for bidirectional encoders (Kushnareva et al., 2021) do not transfer to decoders.

B Implementation and protocol details

Extraction. Models run in float16 on Apple-silicon (MPS) with eager attention. For each example we run one forward pass with attentions and hidden states, average heads, and form D in float64. Ripser (Bauer, 2021) computes 0D/1D barcodes; per-head MSTs use a batched dense Prim’s algorithm (SciPy’s sparse MST drops explicit zero-weight edges, which occur when $A_{u0} \approx 1$). δ_0 uses the closed form of Corollary 1; $\min_s \delta_s$ is computed by brute force over all apexes. Deflated deletes token 0 and renormalizes each row over the remaining causal support. Answer-only restricts D to the answer tokens without renormalization. The Lookback ratio of head h is the mean over answer tokens t of $\bar{a}_{\text{ctx}}/(\bar{a}_{\text{ctx}} + \bar{a}_{\text{new}})$, with $\bar{a}$ the mean attention from t to prompt tokens and to answer tokens up to t . The LLM-Check score of a head is the mean log diagonal attention (the log-determinant of the triangular attention kernel divided by N), summed over heads per layer. Log-probability features are the mean, minimum, and sum of answer-token log-probabilities, mean and maximum

predictive entropy, and mean maximum softmax probability. Hidden states are mean-pooled over answer tokens at layer $\lfloor L/2 \rfloor$.

Prompts. TruthfulQA uses each model’s native format: the Qwen chat template with the default system turn, $\langle |user| \rangle \dots \langle |end| \rangle$ for Phi-3, $\langle |user| \rangle \dots \langle /s \rangle$ for TinyLlama, ChatML without system turn for SmolLM, and $[INST] \dots [/INST]$ for Mistral. HaluEval uses Knowledge: $\dots \backslash n$ Question: $\dots \backslash n$ Answer: $\dots$. The on-policy prompt is the model’s chat template around “Answer the following question with a short phrase only, no explanation. Question: $\dots$ ”. Token 0 is whatever token the format places first (BOS for TinyLlama and Mistral, $\langle |user| \rangle$ for Phi-3, whose tokenizer adds no BOS, $\langle |im_start| \rangle$ for Qwen and SmolLM, and “Knowledge” for Qwen on HaluEval); the theorem does not require it to be a special token.

Table 4: TriviaQA on-policy benchmark. 2,000 validation questions sampled with seed 0; greedy decoding, first line of the answer, at most 24 new tokens. An answer is correct if any normalized gold alias (lower-cased, punctuation and articles removed) occurs in the normalized answer as a whole-word span. Empty answers are dropped.

Model	questions	accuracy	rows used	mean answer tokens
Qwen2.5-3B	2,000	42.4%	2,000	5.5
Qwen2.5-1.5B	2,000	31.4%	2,000	5.8
Phi-3-mini	2,000	56.2%	2,000	6.0
TinyLlama-1.1B	2,000	26.4%	2,000	18.9

On-policy benchmark.

Statistics. Bootstrap resamples are drawn over questions (both rows of a pair move together). TOST power at margin ϵ assumes a true difference of 0 and a normal sampling distribution with the bootstrap SE: $2\Phi(\epsilon/SE - z_{0.95}) - 1$. Repeated-CV predictions are averaged before bootstrapping, so the intervals reflect sampling of questions for a fixed, split-averaged predictor; the seed-to-seed SD of the pooled AUC is reported separately and is below 0.004 everywhere.

C Full results

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA Qw3B	NONTOPO+DEFL	2954	0.928	0.0007	0.931
TQA Qw3B	0D	72	0.754	0.0017	0.799
TQA Qw3B	SINK	72	0.740	0.0028	0.765
TQA Qw3B	1D	144	0.602	0.0018	0.583
TQA Qw3B	DEFL	216	0.717	0.0009	0.748
TQA Qw3B	ANS	216	0.752	0.0017	0.745
TQA Qw3B	ROWSTAT	72	0.751	0.0003	0.829
TQA Qw3B	LOGPROB	6	0.631	0.0008	0.682
TQA Qw3B	HIDDEN	2048	0.913	0.0008	0.916
TQA Qw3B	LEX	–	0.820	0.0014	0.829
TQA Qw3B	LEGACY MSTPROXY	36	0.714	0.0006	0.786
TQA Qw3B	LLMCHECK	36	0.693	0.0011	0.764
TQA Qw3B	LOOKBACK	576	0.897	0.0026	0.906
TQA Qw3B	PH 0D	1152	0.854	0.0022	0.857
TQA Qw3B	PH 0DTOT	576	0.844	0.0010	0.868
TQA Qw3B	PH SINK	576	0.850	0.0018	0.865
TQA Qw3B	PH ENT	576	0.859	0.0016	0.906
TQA Qw3B	0D+SINK	144	0.757	0.0019	0.788
TQA Qw3B	0D+1D	216	0.760	0.0012	0.789
TQA Qw3B	SINK+DEFL	288	0.772	0.0032	0.802
TQA Qw3B	SINK+ANS	288	0.784	0.0026	0.791
TQA Qw3B	LEX+DEFL	–	0.829	0.0010	0.842

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA Qw3B	LEX+SINK	–	0.843	0.0021	0.859
TQA Qw3B	LEX+0D	–	0.847	0.0016	0.860
TQA Qw3B	HIDDEN+0D	2120	0.914	0.0009	0.917
TQA Qw3B	LOGPROB+0D	78	0.787	0.0022	0.820
TQA Qw3B	SINK+1D	216	0.747	0.0021	0.767
TQA Qw3B	NONTOPO+0D	2810	0.928	0.0004	0.936
TQA Qw3B	LEN	2	0.529	0.0003	0.567
TQA Qw3B	NONTOPO	2738	0.927	0.0006	0.938
TQA Qw3B	PH SINK+PH 0DTOT	1152	0.860	0.0032	0.889
TQA Qw1.5B	0D	56	0.726	0.0007	0.767
TQA Qw1.5B	LEN	2	0.529	0.0003	0.567
TQA Qw1.5B	LEX+SINK	–	0.833	0.0008	0.846
TQA Qw1.5B	LEX+0D	–	0.835	0.0017	0.841
TQA Qw1.5B	HIDDEN+0D	1592	0.916	0.0014	0.924
TQA Qw1.5B	LOGPROB+0D	62	0.740	0.0013	0.797
TQA Qw1.5B	NONTOPO+DEFL	2130	0.917	0.0014	0.923
TQA Qw1.5B	NONTOPO+0D	2018	0.917	0.0014	0.927
TQA Qw1.5B	NONTOPO	1962	0.917	0.0013	0.928
TQA Qw1.5B	PH SINK+PH 0DTOT	672	0.841	0.0013	0.867
TQA Qw1.5B	SINK+1D	168	0.715	0.0017	0.769
TQA Qw1.5B	SINK+ANS	224	0.761	0.0027	0.753
TQA Qw1.5B	SINK+DEFL	224	0.757	0.0028	0.802
TQA Qw1.5B	0D+1D	168	0.718	0.0013	0.767
TQA Qw1.5B	0D+SINK	112	0.735	0.0012	0.766
TQA Qw1.5B	LEX+DEFL	–	0.825	0.0019	0.836
TQA Qw1.5B	PH SINK	336	0.830	0.0008	0.851
TQA Qw1.5B	PH ENT	336	0.833	0.0025	0.852
TQA Qw1.5B	1D	112	0.586	0.0014	0.644
TQA Qw1.5B	DEFL	168	0.703	0.0042	0.755
TQA Qw1.5B	ANS	168	0.697	0.0040	0.700
TQA Qw1.5B	ROWSTAT	56	0.738	0.0014	0.815
TQA Qw1.5B	LOGPROB	6	0.638	0.0007	0.682
TQA Qw1.5B	SINK	56	0.723	0.0019	0.771
TQA Qw1.5B	LEX	–	0.820	0.0014	0.829
TQA Qw1.5B	LEGACY MSTPROXY	28	0.698	0.0010	0.786
TQA Qw1.5B	LLMCHECK	28	0.666	0.0005	0.743
TQA Qw1.5B	LOOKBACK	336	0.851	0.0020	0.856
TQA Qw1.5B	PH 0D	672	0.838	0.0012	0.847
TQA Qw1.5B	HIDDEN	1536	0.916	0.0012	0.923
TQA Qw1.5B	PH 0DTOT	336	0.819	0.0011	0.838
TQA Phi-3	LOOKBACK	1024	0.892	0.0004	0.903
TQA Phi-3	LLMCHECK	32	0.653	0.0007	0.754
TQA Phi-3	LEGACY MSTPROXY	32	0.614	0.0008	0.695
TQA Phi-3	LEX	–	0.820	0.0014	0.829
TQA Phi-3	HIDDEN	3072	0.933	0.0007	0.947
TQA Phi-3	ANS	192	0.623	0.0048	0.655
TQA Phi-3	ROWSTAT	64	0.661	0.0011	0.734
TQA Phi-3	DEFL	192	0.639	0.0011	0.685
TQA Phi-3	1D	128	0.529	0.0021	0.551
TQA Phi-3	SINK	64	0.652	0.0007	0.707
TQA Phi-3	PH 0D	2048	0.868	0.0006	0.879
TQA Phi-3	LOGPROB	6	0.578	0.0017	0.599
TQA Phi-3	PH 0DTOT	1024	0.846	0.0020	0.884
TQA Phi-3	LEX+DEFL	–	0.802	0.0010	0.827
TQA Phi-3	PH ENT	1024	0.864	0.0011	0.897
TQA Phi-3	0D+SINK	128	0.658	0.0007	0.707
TQA Phi-3	0D+1D	192	0.645	0.0012	0.707
TQA Phi-3	SINK+DEFL	256	0.685	0.0015	0.727
TQA Phi-3	SINK+ANS	256	0.672	0.0013	0.709
TQA Phi-3	SINK+1D	192	0.654	0.0010	0.705
TQA Phi-3	NONTOPO	4198	0.936	0.0006	0.941
TQA Phi-3	NONTOPO+0D	4262	0.936	0.0006	0.944
TQA Phi-3	NONTOPO+DEFL	4390	0.937	0.0007	0.942

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA Phi-3	LOGPROB+0D	70	0.656	0.0010	0.699
TQA Phi-3	HIDDEN+0D	3136	0.933	0.0008	0.944
TQA Phi-3	LEX+0D	–	0.821	0.0003	0.842
TQA Phi-3	LEX+SINK	–	0.820	0.0005	0.832
TQA Phi-3	PH SINK	1024	0.871	0.0008	0.898
TQA Phi-3	PH SINK+PH 0DTOT	2048	0.871	0.0007	0.906
TQA Phi-3	LEN	2	0.522	0.0004	0.562
TQA Phi-3	0D	64	0.647	0.0010	0.717
TQA TinyLl.	LEN	2	0.523	0.0005	0.556
TQA TinyLl.	0D	44	0.654	0.0012	0.727
TQA TinyLl.	SINK	44	0.644	0.0008	0.720
TQA TinyLl.	1D	88	0.593	0.0010	0.677
TQA TinyLl.	DEFL	132	0.652	0.0028	0.704
TQA TinyLl.	ANS	132	0.646	0.0041	0.683
TQA TinyLl.	ROWSTAT	44	0.660	0.0016	0.744
TQA TinyLl.	LOGPROB	6	0.637	0.0010	0.646
TQA TinyLl.	HIDDEN	2048	0.883	0.0013	0.879
TQA TinyLl.	LEX	–	0.820	0.0014	0.829
TQA TinyLl.	LEGACY MSTPROXY	22	0.627	0.0008	0.738
TQA TinyLl.	LLMCHECK	22	0.614	0.0007	0.701
TQA TinyLl.	LOOKBACK	704	0.850	0.0028	0.846
TQA TinyLl.	PH 0D	1408	0.805	0.0022	0.821
TQA TinyLl.	LEX+DEFL	–	0.824	0.0011	0.837
TQA TinyLl.	PH SINK+PH 0DTOT	1408	0.811	0.0027	0.840
TQA TinyLl.	LEX+0D	–	0.829	0.0011	0.837
TQA TinyLl.	HIDDEN+0D	2092	0.883	0.0014	0.879
TQA TinyLl.	LOGPROB+0D	50	0.682	0.0010	0.717
TQA TinyLl.	NONTOPO+DEFL	2956	0.884	0.0009	0.882
TQA TinyLl.	NONTOPO+0D	2868	0.885	0.0012	0.887
TQA TinyLl.	NONTOPO	2824	0.885	0.0011	0.887
TQA TinyLl.	PH SINK	704	0.807	0.0025	0.845
TQA TinyLl.	PH 0DTOT	704	0.804	0.0018	0.831
TQA TinyLl.	LEX+SINK	–	0.831	0.0012	0.832
TQA TinyLl.	SINK+ANS	176	0.687	0.0033	0.722
TQA TinyLl.	SINK+DEFL	176	0.689	0.0015	0.750
TQA TinyLl.	0D+1D	132	0.659	0.0017	0.734
TQA TinyLl.	0D+SINK	88	0.663	0.0004	0.723
TQA TinyLl.	PH ENT	704	0.817	0.0009	0.838
TQA TinyLl.	SINK+1D	132	0.651	0.0023	0.741
TQA SmolLM	LEX+DEFL	–	0.822	0.0005	0.842
TQA SmolLM	LEN	2	0.524	0.0003	0.554
TQA SmolLM	NONTOPO+0D	2942	0.899	0.0012	0.902
TQA SmolLM	PH SINK+PH 0DTOT	1536	0.825	0.0021	0.856
TQA SmolLM	NONTOPO	2894	0.899	0.0015	0.903
TQA SmolLM	SINK+DEFL	192	0.652	0.0027	0.674
TQA SmolLM	NONTOPO+DEFL	3038	0.899	0.0011	0.897
TQA SmolLM	LOGPROB+0D	54	0.614	0.0011	0.628
TQA SmolLM	HIDDEN+0D	2096	0.894	0.0021	0.891
TQA SmolLM	LEX+0D	–	0.827	0.0009	0.840
TQA SmolLM	LEX+SINK	–	0.819	0.0015	0.825
TQA SmolLM	0D+1D	144	0.588	0.0018	0.646
TQA SmolLM	0D+SINK	96	0.641	0.0027	0.670
TQA SmolLM	PH ENT	768	0.834	0.0007	0.843
TQA SmolLM	PH SINK	768	0.720	0.0009	0.776
TQA SmolLM	SINK+1D	144	0.632	0.0028	0.645
TQA SmolLM	PH 0DTOT	768	0.819	0.0018	0.846
TQA SmolLM	LOOKBACK	768	0.863	0.0017	0.865
TQA SmolLM	LLMCHECK	24	0.604	0.0006	0.685
TQA SmolLM	LEGACY MSTPROXY	24	0.585	0.0011	0.681
TQA SmolLM	LEX	–	0.820	0.0014	0.829
TQA SmolLM	HIDDEN	2048	0.894	0.0022	0.891
TQA SmolLM	LOGPROB	6	0.589	0.0010	0.591
TQA SmolLM	ROWSTAT	48	0.673	0.0013	0.742

Setting	Bank	dim	AUC	seed SD	pair acc.
TQA SmolLM	ANS	144	0.629	0.0034	0.634
TQA SmolLM	DEFL	144	0.592	0.0024	0.656
TQA SmolLM	1D	96	0.545	0.0006	0.569
TQA SmolLM	SINK	48	0.611	0.0020	0.632
TQA SmolLM	0D	48	0.584	0.0028	0.655
TQA SmolLM	PH 0D	1536	0.798	0.0011	0.820
TQA SmolLM	SINK+ANS	192	0.646	0.0024	0.655
HE Qw3B	LOGPROB+0D	78	0.993	0.0003	0.990
HE Qw3B	LEX+DEFL	–	0.970	0.0005	0.977
HE Qw3B	SINK+1D	216	0.842	0.0004	0.938
HE Qw3B	SINK+ANS	288	0.978	0.0007	0.977
HE Qw3B	SINK+DEFL	288	0.882	0.0026	0.961
HE Qw3B	0D+1D	216	0.830	0.0023	0.944
HE Qw3B	0D+SINK	144	0.858	0.0009	0.946
HE Qw3B	LEGACY MSTPROXY	36	0.733	0.0001	0.960
HE Qw3B	LEX+0D	–	0.979	0.0001	0.983
HE Qw3B	HIDDEN	2048	0.999	0.0000	0.999
HE Qw3B	LOGPROB	6	0.993	0.0003	0.986
HE Qw3B	ROWSTAT	72	0.830	0.0003	0.975
HE Qw3B	ANS	216	0.974	0.0008	0.974
HE Qw3B	DEFL	216	0.832	0.0001	0.958
HE Qw3B	1D	144	0.709	0.0010	0.892
HE Qw3B	SINK	72	0.827	0.0008	0.930
HE Qw3B	0D	72	0.816	0.0009	0.929
HE Qw3B	LEN	2	0.962	0.0001	0.970
HE Qw3B	LEX+SINK	–	0.980	0.0002	0.983
HE Qw3B	HIDDEN+0D	2120	0.999	0.0001	0.999
HE Qw3B	LEX	–	0.981	0.0004	0.977
HE Qw1.5B	LEN	2	0.967	0.0002	0.976
HE Qw1.5B	LEX+DEFL	–	0.956	0.0002	0.986
HE Qw1.5B	LEX+0D	–	0.971	0.0004	0.986
HE Qw1.5B	HIDDEN+0D	1592	0.999	0.0001	1.000
HE Qw1.5B	LOGPROB+0D	62	0.993	0.0004	0.994
HE Qw1.5B	NONTOPO+DEFL	2130	0.999	0.0004	1.000
HE Qw1.5B	NONTOPO+0D	2018	0.999	0.0001	1.000
HE Qw1.5B	NONTOPO	1962	0.999	0.0003	1.000
HE Qw1.5B	PH SINK+PH 0DTOT	672	0.931	0.0024	0.998
HE Qw1.5B	SINK+1D	168	0.826	0.0035	0.940
HE Qw1.5B	SINK+ANS	224	0.970	0.0022	0.990
HE Qw1.5B	SINK+DEFL	224	0.872	0.0028	0.962
HE Qw1.5B	0D+1D	168	0.815	0.0032	0.922
HE Qw1.5B	0D+SINK	112	0.832	0.0015	0.924
HE Qw1.5B	PH ENT	336	0.933	0.0011	0.996
HE Qw1.5B	PH SINK	336	0.922	0.0007	0.998
HE Qw1.5B	LEX+SINK	–	0.968	0.0005	0.984
HE Qw1.5B	PH 0D	672	0.949	0.0027	0.974
HE Qw1.5B	0D	56	0.789	0.0014	0.918
HE Qw1.5B	SINK	56	0.820	0.0015	0.934
HE Qw1.5B	1D	112	0.747	0.0006	0.878
HE Qw1.5B	DEFL	168	0.825	0.0021	0.952
HE Qw1.5B	ANS	168	0.969	0.0010	0.983
HE Qw1.5B	ROWSTAT	56	0.788	0.0017	0.984
HE Qw1.5B	PH 0DTOT	336	0.921	0.0007	0.996
HE Qw1.5B	HIDDEN	1536	0.999	0.0001	1.000
HE Qw1.5B	LEX	–	0.979	0.0004	0.980
HE Qw1.5B	LEGACY MSTPROXY	28	0.719	0.0003	0.976
HE Qw1.5B	LLMCHECK	28	0.791	0.0005	0.978
HE Qw1.5B	LOOKBACK	336	0.997	0.0001	0.996
HE Qw1.5B	LOGPROB	6	0.993	0.0001	0.992
TrQA Qw3B	LOOKBACK	576	0.861	0.0013	–
TrQA Qw3B	LLMCHECK	36	0.688	0.0012	–
TrQA Qw3B	LEGACY MSTPROXY	36	0.691	0.0033	–
TrQA Qw3B	LEX	–	0.596	0.0039	–

Setting	Bank	dim	AUC	seed SD	pair acc.
TrQA Qw3B	HIDDEN	2048	0.841	0.0042	—
TrQA Qw3B	LOGPROB	6	0.839	0.0002	—
TrQA Qw3B	DEFL	216	0.684	0.0043	—
TrQA Qw3B	ANS	216	0.719	0.0020	—
TrQA Qw3B	1D	144	0.601	0.0031	—
TrQA Qw3B	SINK	72	0.717	0.0038	—
TrQA Qw3B	0D	72	0.732	0.0038	—
TrQA Qw3B	PH 0D	1152	0.807	0.0017	—
TrQA Qw3B	ROWSTAT	72	0.724	0.0046	—
TrQA Qw3B	LEN	2	0.569	0.0014	—
TrQA Qw3B	PH 0DTOT	576	0.807	0.0012	—
TrQA Qw3B	PH ENT	576	0.810	0.0007	—
TrQA Qw3B	PH SINK	576	0.820	0.0021	—
TrQA Qw3B	LEX+SINK	—	0.724	0.0005	—
TrQA Qw3B	LEX+0D	—	0.733	0.0011	—
TrQA Qw3B	HIDDEN+0D	2120	0.843	0.0040	—
TrQA Qw3B	LOGPROB+0D	78	0.843	0.0022	—
TrQA Qw3B	NONTOPO+DEFL	2954	0.870	0.0029	—
TrQA Qw3B	NONTOPO+0D	2810	0.870	0.0027	—
TrQA Qw3B	LEX+DEFL	—	0.690	0.0031	—
TrQA Qw3B	PH SINK+PH 0DTOT	1152	0.824	0.0017	—
TrQA Qw3B	SINK+1D	216	0.718	0.0035	—
TrQA Qw3B	SINK+ANS	288	0.752	0.0010	—
TrQA Qw3B	SINK+DEFL	288	0.719	0.0038	—
TrQA Qw3B	0D+1D	216	0.729	0.0022	—
TrQA Qw3B	0D+SINK	144	0.738	0.0050	—
TrQA Qw3B	NONTOPO	2738	0.870	0.0028	—
TrQA Qw1.5B	LOOKBACK	336	0.824	0.0017	—
TrQA Qw1.5B	LLMCHECK	28	0.676	0.0029	—
TrQA Qw1.5B	LEGACY MSTPROXY	28	0.685	0.0014	—
TrQA Qw1.5B	LEX	—	0.609	0.0033	—
TrQA Qw1.5B	HIDDEN	1536	0.855	0.0031	—
TrQA Qw1.5B	LOGPROB	6	0.823	0.0006	—
TrQA Qw1.5B	PH 0D	672	0.768	0.0013	—
TrQA Qw1.5B	ANS	168	0.675	0.0035	—
TrQA Qw1.5B	1D	112	0.574	0.0024	—
TrQA Qw1.5B	SINK	56	0.707	0.0000	—
TrQA Qw1.5B	0D	56	0.716	0.0004	—
TrQA Qw1.5B	LEN	2	0.588	0.0012	—
TrQA Qw1.5B	ROWSTAT	56	0.720	0.0008	—
TrQA Qw1.5B	DEFL	168	0.655	0.0016	—
TrQA Qw1.5B	PH 0DTOT	336	0.759	0.0051	—
TrQA Qw1.5B	PH ENT	336	0.768	0.0025	—
TrQA Qw1.5B	PH SINK	336	0.767	0.0034	—
TrQA Qw1.5B	LEX+DEFL	—	0.672	0.0047	—
TrQA Qw1.5B	LEX+0D	—	0.721	0.0031	—
TrQA Qw1.5B	HIDDEN+0D	1592	0.855	0.0032	—
TrQA Qw1.5B	LOGPROB+0D	62	0.828	0.0007	—
TrQA Qw1.5B	NONTOPO+DEFL	2130	0.872	0.0024	—
TrQA Qw1.5B	NONTOPO+0D	2018	0.873	0.0025	—
TrQA Qw1.5B	LEX+SINK	—	0.715	0.0036	—
TrQA Qw1.5B	PH SINK+PH 0DTOT	672	0.772	0.0022	—
TrQA Qw1.5B	SINK+1D	168	0.709	0.0028	—
TrQA Qw1.5B	SINK+ANS	224	0.726	0.0059	—
TrQA Qw1.5B	SINK+DEFL	224	0.719	0.0023	—
TrQA Qw1.5B	NONTOPO	1962	0.873	0.0023	—
TrQA Qw1.5B	0D+1D	168	0.714	0.0015	—
TrQA Qw1.5B	0D+SINK	112	0.717	0.0006	—
TrQA Phi-3	LOOKBACK	1024	0.840	0.0016	—
TrQA Phi-3	LLMCHECK	32	0.676	0.0018	—
TrQA Phi-3	LEGACY MSTPROXY	32	0.669	0.0002	—
TrQA Phi-3	LEX	—	0.635	0.0083	—
TrQA Phi-3	HIDDEN	3072	0.867	0.0035	—

Setting	Bank	dim	AUC	seed SD	pair acc.
TrQA Phi-3	ROWSTAT	64	0.728	0.0006	—
TrQA Phi-3	PH 0D	2048	0.807	0.0024	—
TrQA Phi-3	ANS	192	0.710	0.0037	—
TrQA Phi-3	DEFL	192	0.698	0.0021	—
TrQA Phi-3	1D	128	0.546	0.0031	—
TrQA Phi-3	SINK	64	0.707	0.0018	—
TrQA Phi-3	0D	64	0.710	0.0022	—
TrQA Phi-3	LOGPROB	6	0.877	0.0002	—
TrQA Phi-3	PH 0DTOT	1024	0.810	0.0016	—
TrQA Phi-3	0D+1D	192	0.705	0.0011	—
TrQA Phi-3	PH ENT	1024	0.817	0.0021	—
TrQA Phi-3	0D+SINK	128	0.713	0.0024	—
TrQA Phi-3	SINK+DEFL	256	0.736	0.0016	—
TrQA Phi-3	SINK+ANS	256	0.736	0.0046	—
TrQA Phi-3	SINK+1D	192	0.699	0.0012	—
TrQA Phi-3	NONTOPO	4198	0.883	0.0030	—
TrQA Phi-3	NONTOPO+0D	4262	0.883	0.0030	—
TrQA Phi-3	NONTOPO+DEFL	4390	0.885	0.0030	—
TrQA Phi-3	LOGPROB+0D	70	0.878	0.0013	—
TrQA Phi-3	HIDDEN+0D	3136	0.867	0.0037	—
TrQA Phi-3	LEX+0D	—	0.721	0.0031	—
TrQA Phi-3	LEX+SINK	—	0.718	0.0044	—
TrQA Phi-3	LEX+DEFL	—	0.712	0.0022	—
TrQA Phi-3	LEN	2	0.581	0.0022	—
TrQA Phi-3	PH SINK	1024	0.826	0.0010	—
TrQA Phi-3	PH SINK+PH 0DTOT	2048	0.828	0.0009	—
TrQA TinyLl.	LEX+SINK	—	0.762	0.0021	—
TrQA TinyLl.	LEN	2	0.631	0.0005	—
TrQA TinyLl.	LEX+0D	—	0.764	0.0010	—
TrQA TinyLl.	HIDDEN+0D	2092	0.828	0.0027	—
TrQA TinyLl.	LOGPROB+0D	50	0.762	0.0021	—
TrQA TinyLl.	NONTOPO+DEFL	2956	0.842	0.0032	—
TrQA TinyLl.	NONTOPO+0D	2868	0.842	0.0029	—
TrQA TinyLl.	NONTOPO	2824	0.842	0.0031	—
TrQA TinyLl.	PH SINK+PH 0DTOT	1408	0.813	0.0016	—
TrQA TinyLl.	SINK+1D	132	0.719	0.0013	—
TrQA TinyLl.	SINK+DEFL	176	0.734	0.0032	—
TrQA TinyLl.	0D+1D	132	0.718	0.0002	—
TrQA TinyLl.	0D+SINK	88	0.724	0.0015	—
TrQA TinyLl.	PH ENT	704	0.807	0.0003	—
TrQA TinyLl.	PH SINK	704	0.814	0.0016	—
TrQA TinyLl.	PH 0DTOT	704	0.792	0.0018	—
TrQA TinyLl.	PH 0D	1408	0.775	0.0027	—
TrQA TinyLl.	LOOKBACK	704	0.823	0.0024	—
TrQA TinyLl.	LLMCHECK	22	0.688	0.0010	—
TrQA TinyLl.	LEGACY MSTPROXY	22	0.729	0.0006	—
TrQA TinyLl.	LEX	—	0.740	0.0003	—
TrQA TinyLl.	HIDDEN	2048	0.827	0.0028	—
TrQA TinyLl.	LOGPROB	6	0.724	0.0006	—
TrQA TinyLl.	ROWSTAT	44	0.752	0.0001	—
TrQA TinyLl.	ANS	132	0.701	0.0021	—
TrQA TinyLl.	DEFL	132	0.726	0.0024	—
TrQA TinyLl.	1D	88	0.659	0.0007	—
TrQA TinyLl.	SINK	44	0.720	0.0020	—
TrQA TinyLl.	0D	44	0.720	0.0020	—
TrQA TinyLl.	LEX+DEFL	—	0.757	0.0032	—
TrQA TinyLl.	SINK+ANS	176	0.725	0.0030	—

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI $\equiv_{.015}$	power
TQA Qw3B	LEX $\rightarrow$ LEX+0D	0.820	0.847	+0.027	[+0.016, +0.038]	no 0.48
TQA Qw3B	0D $\rightarrow$ LEGACY_MSTPROXY	0.754	0.714	-0.041	[-0.053, -0.028]	no 0.26
TQA Qw3B	SINK $\rightarrow$ 0D+SINK	0.740	0.757	+0.017	[+0.009, +0.026]	no 0.81

	Setting	Test (A → B)	AUC _A	AUC _B	Δ	90% CI	≡.015	power
1026	TQA Qw3B	SINK→SINK+DEFL	0.740	0.772	+0.032	[+0.019, +0.044]	no	0.31
1027	TQA Qw3B	0D→0D+1D	0.754	0.760	+0.006	[-0.002, +0.013]	yes	0.89
1028	TQA Qw3B	SINK→SINK+1D	0.740	0.747	+0.007	[-0.001, +0.014]	yes	0.90
1029	TQA Qw3B	SINK→SINK+ANS	0.740	0.784	+0.044	[+0.029, +0.059]	no	0.04
1030	TQA Qw3B	PH_0DTOT→PH_SINK	0.844	0.850	+0.007	[-0.001, +0.014]	yes	0.92
1031	TQA Qw3B	PH_SINK→PH_SINK+PH_0DTOT	0.850	0.860	+0.010	[+0.006, +0.014]	yes	1.00
1032	TQA Qw3B	PH_0D→PH_ENT	0.854	0.859	+0.005	[-0.005, +0.015]	no	0.55
1033	TQA Qw3B	0D→SINK	0.754	0.740	-0.014	[-0.024, -0.004]	no	0.59
1034	TQA Qw3B	0D→LOGPROB	0.754	0.631	-0.124	[-0.146, -0.101]	no	0.00
1035	TQA Qw3B	0D→LOOKBACK	0.754	0.897	+0.143	[+0.128, +0.157]	no	0.03
1036	TQA Qw3B	0D→LLMCHECK	0.754	0.693	-0.061	[-0.078, -0.044]	no	0.00
1037	TQA Qw3B	0D→LEX	0.754	0.820	+0.066	[+0.047, +0.084]	no	0.00
1038	TQA Qw3B	0D→ROWSTAT	0.754	0.751	-0.003	[-0.016, +0.010]	no	0.22
1039	TQA Qw3B	0D→LEN	0.754	0.529	-0.226	[-0.248, -0.203]	no	0.00
1040	TQA Qw3B	HIDDEN→HIDDEN+0D	0.913	0.914	+0.001	[-0.000, +0.002]	yes	1.00
1041	TQA Qw3B	LOGPROB→LOGPROB+0D	0.631	0.787	+0.157	[+0.138, +0.176]	no	0.00
1042	TQA Qw3B	LEX→LEX+DEFL	0.820	0.829	+0.009	[-0.004, +0.022]	no	0.22
1043	TQA Qw3B	LEX→LEX+SINK	0.820	0.843	+0.023	[+0.013, +0.033]	no	0.57
1044	TQA Qw3B	0D→HIDDEN	0.754	0.913	+0.159	[+0.144, +0.174]	no	0.03
1045	TQA Qw3B	NONTOPO→NONTOPO+0D	0.927	0.928	+0.001	[-0.000, +0.001]	yes	1.00
1046	TQA Qw3B	NONTOPO→NONTOPO+DEFL	0.927	0.928	+0.001	[-0.000, +0.003]	yes	1.00
1047	TQA Qw1.5B	0D→LEGACY_MSTPROXY	0.726	0.698	-0.028	[-0.040, -0.016]	no	0.36
1048	TQA Qw1.5B	0D→SINK	0.726	0.723	-0.003	[-0.012, +0.006]	yes	0.68
1049	TQA Qw1.5B	LOGPROB→LOGPROB+0D	0.638	0.740	+0.101	[+0.086, +0.116]	no	0.00
1050	TQA Qw1.5B	LEX→LEX+DEFL	0.820	0.825	+0.005	[-0.006, +0.016]	no	0.45
1051	TQA Qw1.5B	LEX→LEX+SINK	0.820	0.833	+0.013	[+0.004, +0.022]	no	0.73
1052	TQA Qw1.5B	LEX→LEX+0D	0.820	0.835	+0.015	[+0.006, +0.024]	no	0.71
1053	TQA Qw1.5B	NONTOPO→NONTOPO+DEFL	0.917	0.917	+0.000	[-0.001, +0.001]	yes	1.00
1054	TQA Qw1.5B	NONTOPO→NONTOPO+0D	0.917	0.917	+0.000	[-0.000, +0.001]	yes	1.00
1055	TQA Qw1.5B	0D→LEN	0.726	0.529	-0.197	[-0.218, -0.176]	no	0.00
1056	TQA Qw1.5B	0D→ROWSTAT	0.726	0.738	+0.012	[-0.000, +0.025]	no	0.27
1057	TQA Qw1.5B	0D→LEX	0.726	0.820	+0.094	[+0.076, +0.112]	no	0.00
1058	TQA Qw1.5B	0D→LLMCHECK	0.726	0.666	-0.060	[-0.076, -0.044]	no	0.00
1059	TQA Qw1.5B	HIDDEN→HIDDEN+0D	0.916	0.916	+0.001	[-0.000, +0.002]	yes	1.00
1060	TQA Qw1.5B	0D→HIDDEN	0.726	0.916	+0.190	[+0.175, +0.204]	no	0.05
1061	TQA Qw1.5B	0D→LOGPROB	0.726	0.638	-0.088	[-0.108, -0.067]	no	0.00
1062	TQA Qw1.5B	PH_0D→PH_ENT	0.838	0.833	-0.004	[-0.016, +0.007]	no	0.44
1063	TQA Qw1.5B	PH_SINK→PH_SINK+PH_0DTOT	0.830	0.841	+0.011	[+0.007, +0.014]	yes	1.00
1064	TQA Qw1.5B	PH_0DTOT→PH_SINK	0.819	0.830	+0.011	[+0.003, +0.018]	no	0.90
1065	TQA Qw1.5B	SINK→SINK+ANS	0.723	0.761	+0.038	[+0.024, +0.051]	no	0.11
1066	TQA Qw1.5B	SINK→SINK+1D	0.723	0.715	-0.008	[-0.014, -0.002]	yes	0.99
1067	TQA Qw1.5B	0D→0D+1D	0.726	0.718	-0.008	[-0.014, -0.002]	yes	0.99
1068	TQA Qw1.5B	SINK→SINK+DEFL	0.723	0.757	+0.034	[+0.022, +0.045]	no	0.35
1069	TQA Qw1.5B	SINK→0D+SINK	0.723	0.735	+0.012	[+0.005, +0.020]	no	0.91
1070	TQA Qw1.5B	0D→LOOKBACK	0.726	0.851	+0.125	[+0.109, +0.140]	no	0.00
1071	TQA Phi-3	PH_0D→PH_ENT	0.868	0.864	-0.004	[-0.012, +0.004]	yes	0.87
1072	TQA Phi-3	PH_SINK→PH_SINK+PH_0DTOT	0.871	0.871	+0.000	[-0.001, +0.001]	yes	1.00
1073	TQA Phi-3	PH_0DTOT→PH_SINK	0.846	0.871	+0.025	[+0.020, +0.030]	no	1.00
1074	TQA Phi-3	SINK→0D+SINK	0.652	0.658	+0.006	[+0.000, +0.012]	yes	0.99
1075	TQA Phi-3	SINK→SINK+1D	0.652	0.654	+0.002	[-0.003, +0.006]	yes	1.00
1076	TQA Phi-3	0D→0D+1D	0.647	0.645	-0.001	[-0.005, +0.002]	yes	1.00
1077	TQA Phi-3	SINK→SINK+DEFL	0.652	0.685	+0.033	[+0.019, +0.047]	no	0.08
1078	TQA Phi-3	0D→LOGPROB	0.647	0.578	-0.069	[-0.093, -0.045]	no	0.00
1079	TQA Phi-3	SINK→SINK+ANS	0.652	0.672	+0.020	[+0.004, +0.036]	no	0.00
1080	TQA Phi-3	0D→HIDDEN	0.647	0.933	+0.286	[+0.271, +0.301]	no	0.00
1081	TQA Phi-3	HIDDEN→HIDDEN+0D	0.933	0.933	+0.000	[-0.000, +0.001]	yes	1.00
1082	TQA Phi-3	0D→LLMCHECK	0.647	0.653	+0.007	[-0.011, +0.024]	no	0.00
1083	TQA Phi-3	0D→LEX	0.647	0.820	+0.173	[+0.153, +0.193]	no	0.00
1084	TQA Phi-3	0D→ROWSTAT	0.647	0.661	+0.014	[+0.001, +0.028]	no	0.12
1085	TQA Phi-3	NONTOPO→NONTOPO+0D	0.936	0.936	+0.000	[-0.000, +0.000]	yes	1.00
1086	TQA Phi-3	NONTOPO→NONTOPO+DEFL	0.936	0.937	+0.001	[-0.000, +0.002]	yes	1.00
1087	TQA Phi-3	LEX→LEX+0D	0.820	0.821	+0.001	[-0.007, +0.009]	yes	0.85
1088	TQA Phi-3	LEX→LEX+SINK	0.820	0.820	-0.000	[-0.008, +0.008]	yes	0.88
1089	TQA Phi-3	LEX→LEX+DEFL	0.820	0.802	-0.018	[-0.028, -0.007]	no	0.52
1090	TQA Phi-3	LOGPROB→LOGPROB+0D	0.578	0.656	+0.078	[+0.059, +0.097]	no	0.00
1091	TQA Phi-3	0D→LOOKBACK	0.647	0.892	+0.245	[+0.229, +0.261]	no	0.00
1092	TQA Phi-3	0D→LEN	0.647	0.522	-0.124	[-0.146, -0.103]	no	0.00
1093	TQA Phi-3	0D→SINK	0.647	0.652	+0.005	[-0.002, +0.013]	yes	0.91
1094	TQA Phi-3	0D→LEGACY_MSTPROXY	0.647	0.614	-0.033	[-0.046, -0.020]	no	0.20
1095	TQA TinyLl.	0D→SINK	0.654	0.644	-0.010	[-0.021, +0.001]	no	0.48
1096	TQA TinyLl.	0D→LEGACY_MSTPROXY	0.654	0.627	-0.027	[-0.038, -0.016]	no	0.45
1097	TQA TinyLl.	SINK→0D+SINK	0.644	0.663	+0.020	[+0.010, +0.029]	no	0.72
1098	TQA TinyLl.	SINK→SINK+DEFL	0.644	0.689	+0.045	[+0.033, +0.058]	no	0.25
1099	TQA TinyLl.	0D→0D+1D	0.654	0.659	+0.005	[-0.002, +0.012]	yes	0.94
1100	TQA TinyLl.	SINK→SINK+1D	0.644	0.651	+0.007	[-0.000, +0.015]	yes	0.90
1101	TQA TinyLl.	SINK→SINK+ANS	0.644	0.687	+0.043	[+0.026, +0.060]	no	0.00
1102	TQA TinyLl.	PH_0DTOT→PH_SINK	0.804	0.807	+0.003	[-0.003, +0.009]	yes	0.97
1103	TQA TinyLl.	PH_SINK→PH_SINK+PH_0DTOT	0.807	0.811	+0.004	[+0.002, +0.005]	yes	1.00
1104	TQA TinyLl.	PH_0D→PH_ENT	0.805	0.817	+0.011	[+0.001, +0.022]	no	0.50
1105	TQA TinyLl.	0D→LOGPROB	0.654	0.637	-0.017	[-0.039, +0.004]	no	0.00
1106	TQA TinyLl.	HIDDEN→HIDDEN+0D	0.883	0.883	-0.000	[-0.001, +0.000]	yes	1.00
1107	TQA TinyLl.	0D→LOOKBACK	0.654	0.850	+0.196	[+0.182, +0.212]	no	0.00
1108	TQA TinyLl.	0D→LLMCHECK	0.654	0.614	-0.040	[-0.055, -0.024]	no	0.00
1109	TQA TinyLl.	0D→LEX	0.654	0.820	+0.166	[+0.149, +0.184]	no	0.00
1110	TQA TinyLl.	0D→ROWSTAT	0.654	0.660	+0.006	[-0.005, +0.018]	no	0.37
1111	TQA TinyLl.	LOGPROB→LOGPROB+0D	0.637	0.682	+0.046	[+0.032, +0.059]	no	0.13
1112	TQA TinyLl.	0D→LEN	0.654	0.523	-0.131	[-0.152, -0.110]	no	0.00
1113	TQA TinyLl.	NONTOPO→NONTOPO+0D	0.885	0.885	+0.000	[-0.000, +0.000]	yes	1.00
1114	TQA TinyLl.	NONTOPO→NONTOPO+DEFL	0.885	0.884	-0.001	[-0.002, -0.000]	yes	1.00
1115	TQA TinyLl.	LEX→LEX+0D	0.820	0.829	+0.009	[+0.003, +0.016]	no	0.98

	Setting	Test (A → B)	AUC _A	AUC _B	Δ	90% CI	≡.015	power
1080	TQA TinyLl.	LEX→LEX+SINK	0.820	0.831	+0.011	[+0.005, +0.017]	no	0.98
1081	TQA TinyLl.	LEX→LEX+DEFL	0.820	0.824	+0.004	[-0.005, +0.013]	yes	0.75
1082	TQA TinyLl.	0D→HIDDEN	0.654	0.883	+0.229	[+0.213, +0.244]	no	0.00
1083	TQA SmolLM	HIDDEN→HIDDEN+0D	0.894	0.894	+0.000	[-0.001, +0.001]	yes	1.00
1084	TQA SmolLM	0D→SINK	0.584	0.611	+0.027	[+0.008, +0.046]	no	0.00
1085	TQA SmolLM	NONTOPO→NONTOPO+DEFL	0.899	0.899	-0.000	[-0.001, +0.001]	yes	1.00
1086	TQA SmolLM	0D→ROWSTAT	0.584	0.673	+0.090	[+0.078, +0.101]	no	0.36
1087	TQA SmolLM	LEX→LEX+0D	0.820	0.827	+0.007	[+0.001, +0.012]	yes	0.99
1088	TQA SmolLM	LEX→LEX+SINK	0.820	0.819	-0.001	[-0.007, +0.004]	yes	1.00
1089	TQA SmolLM	LEX→LEX+DEFL	0.820	0.822	+0.002	[-0.006, +0.009]	yes	0.90
1090	TQA SmolLM	LOGPROB→LOGPROB+0D	0.589	0.614	+0.025	[+0.015, +0.035]	no	0.58
1091	TQA SmolLM	0D→LEX	0.584	0.820	+0.236	[+0.217, +0.255]	no	0.00
1092	TQA SmolLM	0D→LLMCHECK	0.584	0.604	+0.020	[+0.005, +0.035]	no	0.00
1093	TQA SmolLM	0D→LOOKBACK	0.584	0.863	+0.279	[+0.264, +0.295]	no	0.00
1094	TQA SmolLM	0D→HIDDEN	0.584	0.894	+0.310	[+0.295, +0.325]	no	0.02
1095	TQA SmolLM	0D→LOGPROB	0.584	0.589	+0.005	[-0.017, +0.027]	no	0.00
1096	TQA SmolLM	PH_0D→PH_ENT	0.798	0.834	+0.036	[+0.026, +0.046]	no	0.59
1097	TQA SmolLM	PH_SINK→PH_SINK+PH_0DTOT	0.720	0.825	+0.105	[+0.093, +0.118]	no	0.29
1098	TQA SmolLM	PH_0DTOT→PH_SINK	0.819	0.720	-0.099	[-0.113, -0.086]	no	0.16
1099	TQA SmolLM	SINK→SINK+ANS	0.611	0.646	+0.036	[+0.018, +0.053]	no	0.00
1100	TQA SmolLM	SINK→SINK+1D	0.611	0.632	+0.022	[+0.012, +0.032]	no	0.58
1101	TQA SmolLM	0D→0D+1D	0.584	0.588	+0.004	[-0.005, +0.014]	yes	0.66
1102	TQA SmolLM	SINK→SINK+DEFL	0.611	0.652	+0.041	[+0.030, +0.053]	no	0.36
1103	TQA SmolLM	SINK→0D+SINK	0.611	0.641	+0.030	[+0.021, +0.039]	no	0.67
1104	TQA SmolLM	0D→LEGACY_MSTPROXY	0.584	0.585	+0.001	[-0.004, +0.005]	yes	1.00
1105	TQA SmolLM	NONTOPO→NONTOPO+0D	0.899	0.899	-0.000	[-0.001, +0.000]	yes	1.00
1106	TQA SmolLM	0D→LEN	0.584	0.524	-0.060	[-0.075, -0.044]	no	0.00
1107	HE Qw3B	LEX→LEX+0D	0.981	0.979	-0.002	[-0.005, +0.000]	yes	1.00
1108	HE Qw3B	HIDDEN→HIDDEN+0D	0.999	0.999	-0.000	[-0.000, +0.000]	yes	1.00
1109	HE Qw3B	LOGPROB→LOGPROB+0D	0.993	0.993	-0.000	[-0.002, +0.001]	yes	1.00
1110	HE Qw3B	0D→SINK	0.816	0.827	+0.011	[+0.002, +0.021]	no	0.66
1111	HE Qw3B	0D→LEGACY_MSTPROXY	0.816	0.733	-0.083	[-0.094, -0.072]	no	0.45
1112	HE Qw3B	SINK→0D+SINK	0.827	0.858	+0.030	[+0.023, +0.037]	no	0.96
1113	HE Qw3B	SINK→SINK+DEFL	0.827	0.882	+0.055	[+0.046, +0.063]	no	0.75
1114	HE Qw3B	0D→0D+1D	0.816	0.830	+0.013	[+0.007, +0.020]	no	0.96
1115	HE Qw3B	SINK→SINK+1D	0.827	0.842	+0.014	[+0.008, +0.021]	no	0.96
1116	HE Qw3B	SINK→SINK+ANS	0.827	0.978	+0.151	[+0.139, +0.162]	no	0.41
1117	HE Qw3B	LEX→LEX+DEFL	0.981	0.970	-0.011	[-0.015, -0.007]	yes	1.00
1118	HE Qw3B	0D→HIDDEN	0.816	0.999	+0.183	[+0.171, +0.195]	no	0.33
1119	HE Qw3B	0D→LEX	0.816	0.981	+0.165	[+0.153, +0.176]	no	0.38
1120	HE Qw3B	0D→ROWSTAT	0.816	0.830	+0.014	[+0.002, +0.025]	no	0.34
1121	HE Qw3B	0D→LEN	0.816	0.962	+0.146	[+0.134, +0.158]	no	0.37
1122	HE Qw3B	LEX→LEX+SINK	0.981	0.980	-0.001	[-0.004, +0.001]	yes	1.00
1123	HE Qw3B	0D→LOGPROB	0.816	0.993	+0.177	[+0.165, +0.189]	no	0.35
1124	HE Qw1.5B	0D→SINK	0.789	0.820	+0.031	[+0.017, +0.045]	no	0.08
1125	HE Qw1.5B	HIDDEN→HIDDEN+0D	0.999	0.999	+0.000	[-0.000, +0.000]	yes	1.00
1126	HE Qw1.5B	LOGPROB→LOGPROB+0D	0.993	0.993	-0.001	[-0.003, +0.001]	yes	1.00
1127	HE Qw1.5B	LEX→LEX+DEFL	0.979	0.956	-0.022	[-0.030, -0.016]	no	0.94
1128	HE Qw1.5B	LEX→LEX+SINK	0.979	0.968	-0.011	[-0.017, -0.005]	no	0.98
1129	HE Qw1.5B	LEX→LEX+0D	0.979	0.971	-0.008	[-0.014, -0.002]	yes	0.99
1130	HE Qw1.5B	NONTOPO→NONTOPO+DEFL	0.999	0.999	+0.000	[-0.000, +0.000]	yes	1.00
1131	HE Qw1.5B	NONTOPO→NONTOPO+0D	0.999	0.999	+0.000	[+0.000, +0.001]	yes	1.00
1132	HE Qw1.5B	0D→ROWSTAT	0.789	0.788	-0.001	[-0.018, +0.016]	no	0.00
1133	HE Qw1.5B	0D→LEX	0.789	0.979	+0.190	[+0.172, +0.207]	no	0.00
1134	HE Qw1.5B	0D→LLMCHECK	0.789	0.791	+0.002	[-0.017, +0.021]	no	0.00
1135	HE Qw1.5B	0D→LOOKBACK	0.789	0.997	+0.208	[+0.191, +0.225]	no	0.00
1136	HE Qw1.5B	0D→LEN	0.789	0.967	+0.178	[+0.160, +0.196]	no	0.00
1137	HE Qw1.5B	0D→LOGPROB	0.789	0.993	+0.204	[+0.187, +0.221]	no	0.00
1138	HE Qw1.5B	PH_0D→PH_ENT	0.949	0.933	-0.016	[-0.028, -0.004]	no	0.33
1139	HE Qw1.5B	PH_SINK→PH_SINK+PH_0DTOT	0.922	0.931	+0.008	[+0.006, +0.011]	yes	1.00
1140	HE Qw1.5B	PH_0DTOT→PH_SINK	0.921	0.922	+0.002	[-0.009, +0.013]	yes	0.46
1141	HE Qw1.5B	SINK→SINK+ANS	0.820	0.970	+0.151	[+0.133, +0.168]	no	0.00
1142	HE Qw1.5B	SINK→SINK+1D	0.820	0.826	+0.006	[-0.005, +0.017]	no	0.45
1143	HE Qw1.5B	0D→0D+1D	0.789	0.815	+0.026	[+0.013, +0.040]	no	0.17
1144	HE Qw1.5B	SINK→SINK+DEFL	0.820	0.872	+0.052	[+0.038, +0.065]	no	0.17
1145	HE Qw1.5B	SINK→0D+SINK	0.820	0.832	+0.012	[+0.005, +0.019]	no	0.91
1146	HE Qw1.5B	0D→HIDDEN	0.789	0.999	+0.210	[+0.193, +0.227]	no	0.00
1147	HE Qw1.5B	0D→LEGACY_MSTPROXY	0.789	0.719	-0.070	[-0.086, -0.053]	no	0.00
1148	TrQA Qw3B	PH_SINK→PH_SINK+PH_0DTOT	0.820	0.824	+0.004	[+0.002, +0.006]	yes	1.00
1149	TrQA Qw3B	PH_0DTOT→PH_SINK	0.807	0.820	+0.012	[+0.004, +0.021]	no	0.82
1150	TrQA Qw3B	SINK→SINK+ANS	0.717	0.752	+0.035	[+0.019, +0.051]	no	0.00
1151	TrQA Qw3B	SINK→SINK+1D	0.717	0.718	+0.001	[-0.009, +0.010]	yes	0.65
1152	TrQA Qw3B	0D→SINK	0.732	0.717	-0.015	[-0.026, -0.004]	no	0.48
1153	TrQA Qw3B	SINK→SINK+DEFL	0.717	0.719	+0.002	[-0.012, +0.015]	no	0.13
1154	TrQA Qw3B	SINK→0D+SINK	0.717	0.738	+0.021	[+0.012, +0.030]	no	0.76
1155	TrQA Qw3B	0D→LEGACY_MSTPROXY	0.732	0.691	-0.041	[-0.057, -0.025]	no	0.00
1156	TrQA Qw3B	PH_0D→PH_ENT	0.807	0.810	+0.003	[-0.011, +0.017]	no	0.08
1157	TrQA Qw3B	0D→0D+1D	0.732	0.729	-0.003	[-0.012, +0.006]	yes	0.74
1158	TrQA Qw3B	0D→LOGPROB	0.732	0.839	+0.107	[+0.088, +0.125]	no	0.00
1159	TrQA Qw3B	NONTOPO→NONTOPO+0D	0.870	0.870	+0.001	[-0.000, +0.002]	yes	1.00
1160	TrQA Qw3B	0D→LOOKBACK	0.732	0.861	+0.129	[+0.111, +0.147]	no	0.00
1161	TrQA Qw3B	0D→LLMCHECK	0.732	0.688	-0.044	[-0.065, -0.023]	no	0.00
1162	TrQA Qw3B	0D→LEX	0.732	0.596	-0.137	[-0.163, -0.110]	no	0.00
1163	TrQA Qw3B	0D→ROWSTAT	0.732	0.724	-0.009	[-0.025, +0.008]	no	0.00
1164	TrQA Qw3B	0D→LEN	0.732	0.569	-0.163	[-0.189, -0.137]	no	0.00
1165	TrQA Qw3B	NONTOPO→NONTOPO+DEFL	0.870	0.870	+0.000	[-0.001, +0.002]	yes	1.00
1166	TrQA Qw3B	LEX→LEX+0D	0.596	0.733	+0.137	[+0.117, +0.158]	no	0.00
1167	TrQA Qw3B	LEX→LEX+SINK	0.596	0.724	+0.128	[+0.108, +0.149]	no	0.00
1168	TrQA Qw3B	LEX→LEX+DEFL	0.596	0.690	+0.095	[+0.073, +0.116]	no	0.00
1169	TrQA Qw3B	LOGPROB→LOGPROB+0D	0.839	0.843	+0.004	[-0.002, +0.011]	yes	0.95

Setting	Test (A $\rightarrow$ B)	AUC _A	AUC _B	Δ	90% CI	$\equiv_{.015}$	power
TrQA Qw3B	HIDDEN $\rightarrow$ HIDDEN+0D	0.841	0.843	+0.002	[+0.000, +0.004]	yes	1.00
TrQA Qw3B	0D $\rightarrow$ HIDDEN	0.732	0.841	+0.109	[+0.090, +0.129]	no	0.00
TrQA Qw1.5B	PH_0D $\rightarrow$ PH_ENT	0.768	0.768	-0.000	[-0.017, +0.017]	no	0.00
TrQA Qw1.5B	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.767	0.772	+0.004	[+0.001, +0.008]	yes	1.00
TrQA Qw1.5B	PH_0DTOT $\rightarrow$ PH_SINK	0.759	0.767	+0.008	[-0.003, +0.018]	no	0.53
TrQA Qw1.5B	SINK $\rightarrow$ SINK+ANS	0.707	0.726	+0.019	[+0.001, +0.036]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ SINK	0.716	0.707	-0.009	[-0.020, +0.002]	no	0.43
TrQA Qw1.5B	0D $\rightarrow$ 0D+1D	0.716	0.714	-0.002	[-0.011, +0.008]	yes	0.65
TrQA Qw1.5B	SINK $\rightarrow$ SINK+DEFL	0.707	0.719	+0.012	[-0.003, +0.027]	no	0.00
TrQA Qw1.5B	SINK $\rightarrow$ 0D+SINK	0.707	0.717	+0.010	[+0.001, +0.020]	no	0.68
TrQA Qw1.5B	0D $\rightarrow$ LEGACY_MSTPROXY	0.716	0.685	-0.031	[-0.046, -0.015]	no	0.00
TrQA Qw1.5B	SINK $\rightarrow$ SINK+1D	0.707	0.709	+0.002	[-0.008, +0.013]	yes	0.49
TrQA Qw1.5B	0D $\rightarrow$ HIDDEN	0.716	0.855	+0.140	[+0.119, +0.160]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LOGPROB	0.716	0.823	+0.107	[+0.085, +0.129]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LLMCHECK	0.716	0.676	-0.039	[-0.062, -0.016]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LOOKBACK	0.716	0.824	+0.108	[+0.088, +0.129]	no	0.00
TrQA Qw1.5B	HIDDEN $\rightarrow$ HIDDEN+0D	0.855	0.855	+0.000	[-0.001, +0.002]	yes	1.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+DEFL	0.609	0.672	+0.063	[+0.040, +0.085]	no	0.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+SINK	0.609	0.715	+0.106	[+0.085, +0.127]	no	0.00
TrQA Qw1.5B	LEX $\rightarrow$ LEX+0D	0.609	0.721	+0.112	[+0.091, +0.134]	no	0.00
TrQA Qw1.5B	LOGPROB $\rightarrow$ LOGPROB+0D	0.823	0.828	+0.006	[-0.001, +0.013]	yes	0.92
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+0D	0.873	0.873	+0.000	[-0.001, +0.001]	yes	1.00
TrQA Qw1.5B	0D $\rightarrow$ LEN	0.716	0.588	-0.128	[-0.154, -0.102]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ ROWSTAT	0.716	0.720	+0.005	[-0.011, +0.020]	no	0.00
TrQA Qw1.5B	0D $\rightarrow$ LEX	0.716	0.609	-0.106	[-0.136, -0.078]	no	0.00
TrQA Qw1.5B	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.873	0.872	-0.001	[-0.002, +0.001]	yes	1.00
TrQA Phi-3	0D $\rightarrow$ LEGACY_MSTPROXY	0.710	0.669	-0.041	[-0.057, -0.025]	no	0.00
TrQA Phi-3	SINK $\rightarrow$ 0D+SINK	0.707	0.713	+0.006	[+0.000, +0.012]	yes	0.99
TrQA Phi-3	SINK $\rightarrow$ SINK+DEFL	0.707	0.736	+0.029	[+0.013, +0.045]	no	0.00
TrQA Phi-3	0D $\rightarrow$ 0D+1D	0.710	0.705	-0.005	[-0.011, +0.000]	yes	0.99
TrQA Phi-3	SINK $\rightarrow$ SINK+1D	0.707	0.699	-0.008	[-0.013, -0.002]	yes	1.00
TrQA Phi-3	SINK $\rightarrow$ SINK+ANS	0.707	0.736	+0.029	[+0.012, +0.046]	no	0.00
TrQA Phi-3	PH_0DTOT $\rightarrow$ PH_SINK	0.810	0.826	+0.016	[+0.009, +0.022]	no	0.96
TrQA Phi-3	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.826	0.828	+0.002	[-0.001, +0.004]	yes	1.00
TrQA Phi-3	PH_0D $\rightarrow$ PH_ENT	0.807	0.817	+0.010	[-0.002, +0.022]	no	0.31
TrQA Phi-3	0D $\rightarrow$ LOGPROB	0.710	0.877	+0.167	[+0.148, +0.186]	no	0.00
TrQA Phi-3	0D $\rightarrow$ HIDDEN	0.710	0.867	+0.157	[+0.137, +0.176]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LLMCHECK	0.710	0.676	-0.034	[-0.055, -0.014]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LEX	0.710	0.635	-0.076	[-0.102, -0.050]	no	0.00
TrQA Phi-3	0D $\rightarrow$ ROWSTAT	0.710	0.728	+0.018	[+0.002, +0.034]	no	0.00
TrQA Phi-3	0D $\rightarrow$ LEN	0.710	0.581	-0.130	[-0.156, -0.104]	no	0.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.883	0.885	+0.002	[+0.000, +0.004]	yes	1.00
TrQA Phi-3	LEX $\rightarrow$ LEX+0D	0.635	0.721	+0.086	[+0.069, +0.104]	no	0.00
TrQA Phi-3	LEX $\rightarrow$ LEX+SINK	0.635	0.718	+0.083	[+0.066, +0.101]	no	0.00
TrQA Phi-3	LEX $\rightarrow$ LEX+DEFL	0.635	0.712	+0.077	[+0.057, +0.098]	no	0.00
TrQA Phi-3	LOGPROB $\rightarrow$ LOGPROB+0D	0.877	0.878	+0.000	[-0.002, +0.002]	yes	1.00
TrQA Phi-3	HIDDEN $\rightarrow$ HIDDEN+0D	0.867	0.867	-0.000	[-0.001, +0.000]	yes	1.00
TrQA Phi-3	0D $\rightarrow$ SINK	0.710	0.707	-0.004	[-0.010, +0.002]	yes	0.98
TrQA Phi-3	0D $\rightarrow$ LOOKBACK	0.710	0.840	+0.129	[+0.109, +0.150]	no	0.00
TrQA Phi-3	NONTOPO $\rightarrow$ NONTOPO+0D	0.883	0.883	-0.000	[-0.001, +0.000]	yes	1.00
TrQA TinyLl.	LOGPROB $\rightarrow$ LOGPROB+0D	0.724	0.762	+0.038	[+0.025, +0.052]	no	0.17
TrQA TinyLl.	0D $\rightarrow$ SINK	0.720	0.720	+0.001	[-0.010, +0.012]	yes	0.40
TrQA TinyLl.	LEX $\rightarrow$ LEX+DEFL	0.740	0.757	+0.016	[+0.002, +0.031]	no	0.04
TrQA TinyLl.	LEX $\rightarrow$ LEX+SINK	0.740	0.762	+0.021	[+0.010, +0.032]	no	0.45
TrQA TinyLl.	LEX $\rightarrow$ LEX+0D	0.740	0.764	+0.024	[+0.012, +0.036]	no	0.33
TrQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO+DEFL	0.842	0.842	+0.000	[-0.003, +0.003]	yes	1.00
TrQA TinyLl.	NONTOPO $\rightarrow$ NONTOPO+0D	0.842	0.842	-0.000	[-0.001, +0.001]	yes	1.00
TrQA TinyLl.	0D $\rightarrow$ LEN	0.720	0.631	-0.089	[-0.109, -0.068]	no	0.00
TrQA TinyLl.	0D $\rightarrow$ LEX	0.720	0.740	+0.021	[-0.003, +0.044]	no	0.00
TrQA TinyLl.	0D $\rightarrow$ LLMCHECK	0.720	0.688	-0.032	[-0.050, -0.013]	no	0.00
TrQA TinyLl.	0D $\rightarrow$ LOOKBACK	0.720	0.823	+0.104	[+0.087, +0.121]	no	0.00
TrQA TinyLl.	HIDDEN $\rightarrow$ HIDDEN+0D	0.827	0.828	+0.001	[-0.001, +0.002]	yes	1.00
TrQA TinyLl.	0D $\rightarrow$ HIDDEN	0.720	0.827	+0.108	[+0.088, +0.128]	no	0.00
TrQA TinyLl.	PH_0D $\rightarrow$ PH_ENT	0.775	0.807	+0.032	[+0.017, +0.047]	no	0.01
TrQA TinyLl.	PH_SINK $\rightarrow$ PH_SINK+PH_0DTOT	0.814	0.813	-0.001	[-0.002, +0.000]	yes	1.00
TrQA TinyLl.	PH_0DTOT $\rightarrow$ PH_SINK	0.792	0.814	+0.022	[+0.015, +0.029]	no	0.94
TrQA TinyLl.	SINK $\rightarrow$ SINK+ANS	0.720	0.725	+0.005	[-0.011, +0.021]	no	0.00
TrQA TinyLl.	SINK $\rightarrow$ SINK+1D	0.720	0.719	-0.001	[-0.010, +0.008]	yes	0.73
TrQA TinyLl.	0D $\rightarrow$ 0D+1D	0.720	0.718	-0.002	[-0.011, +0.008]	yes	0.60
TrQA TinyLl.	SINK $\rightarrow$ SINK+DEFL	0.720	0.734	+0.014	[-0.000, +0.028]	no	0.10
TrQA TinyLl.	SINK $\rightarrow$ 0D+SINK	0.720	0.724	+0.004	[-0.003, +0.011]	yes	0.93
TrQA TinyLl.	0D $\rightarrow$ LEGACY_MSTPROXY	0.720	0.729	+0.010	[-0.000, +0.020]	no	0.55
TrQA TinyLl.	0D $\rightarrow$ LOGPROB	0.720	0.724	+0.004	[-0.019, +0.027]	no	0.00
TrQA TinyLl.	0D $\rightarrow$ ROWSTAT	0.720	0.752	+0.032	[+0.020, +0.045]	no	0.26

D Mistral-7B

Mistral-7B-Instruct-v0.2 does not fit in the memory of the machine used for the main experiments. The released extractor runs it unchanged on a single 16GB GPU (sinktda/RUN_ON_5080.md). An accuracy-level audit of this model under its [INST] format found that 99.99% of its 1D features are exactly zero and that 0D and 0D+1D banks are equivalent; by Theorem 1(c) this is what exact coning in every graph would produce, but we have not yet verified $\delta_0 = 0$ on its graphs directly.

E Relation to accuracy-level audits, and cost

Audits of attention-graph PH that stop at classification accuracy observe four facts that the theory now explains: (i) 0D persistence equals an MST statistic (Fact 1); (ii) 1D persistence adds little beyond 0D (Table 3, column “1D | 0D”), which Theorem 1 now explains; (iii) the 1D bank of Mistral-7B under its [INST] format is 99.99% zeros, which is exactly the coned regime; (iv) HaluEval answers are separable from text alone.

Cost. Median per-matrix wall-clock time on real attention from TinyLlama-1.1B-Chat-v1.0 (sink features / δ_0 / dense MST / ripser 0D / ripser 0D+1D) is 0.001/0.02/0.10/0.02/0.02 ms at $N=32$; 0.001/0.03/0.21/0.03/0.03 ms at $N=64$; 0.001/0.07/0.42/0.07/0.06 ms at $N=128$; 0.001/0.25/0.90/0.18/0.18 ms at $N=256$. On i.i.d. random matrices of the same size, ripser 0D+1D takes 0.171 ms ($N=32$), 3.46 ms ($N=64$), 115 ms ($N=128$), 4,631 ms ($N=256$). On real attention the same computation is four orders of magnitude cheaper at $N=256$, consistent with Theorem 1: sink-dominated graphs have few, short 1D bars to compute. Sink features, read off one column, are one to two orders of magnitude cheaper still.